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ORIGINAL ARTICLE

Glaucoma detection and severity classification based on glaucoattent net framework

Sachin Chavan[1] · Nitin Choubey[1]

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Abstract

Global blindness due to glaucoma necessitates advanced diagnostics. Automated glaucoma detection models, while introduced, grapple with discerning subtle optic disc (OD) and optic cup (OC) changes. Challenges arise in intricate segmentation due to structural variations. Moreover, many existing works fixate on binary classification, lacking nuanced assessments, highlighting the need for more sophisticated approaches. To tackle these challenges, we introduce a framework called GlaucoAttent Net for glaucoma detection, aiming to effectively segment the OD and OC while assessing severity. The foundation of robust model training lies in diverse datasets—ORIGA, REFUGE and G1020. Initially, a glaucoma feature enhanced pre-processing stage preserves fundus image details through Non-Local Means (NLM) denoising, Contrast Limited Adaptive Histogram Equalization (CLAHE) and Adaptive Median Filtering. It retains fine structures, texture patterns, OD morphology, intensity gradients and localized features essential for segmentation. Within the core engine GlaucoAttent Net, which is a cascaded U-Net, the encoder path efficiently extracts feature at two levels to create hierarchical feature representations essential for segmenting the OD. The decoder employs a novel attention mechanism to create a probability mask for OD segmentation followed by OC segmentation by refining features. At the last stage of decoder, merging the probability masks for accurate delineation of the OD and OC boundaries in the image. Additionally, a classification system analyses segmentation outputs like cup-to-disc ratio, areas, diameters and other features to categorize eyes as Healthy, Highly Glaucomatous, Less Glaucomatous and Moderately Glaucomatous. Efficacy metrics showcase outstanding performance: 99.67% accuracy, 99.20% precision, 99.50% recall and a 99.35% F-score, solidifying the methodology's potential for transformative impact in advancing clinical diagnostics**.**

Keywords Deep learning · GlaucoAttent Net · Glaucoma detection · OD segmentation · OC Segmentation · Severity assessment

1 Introduction

Glaucoma, a major cause of permanent blindness, progressively impairs the optic nerve, primarily affecting structures like the optic disc (OD) and optic cup (OC) [1]. Monitoring changes in the cup-to-disc ratio (CDR) is crucial for diagnosing and tracking glaucoma progression. However, traditional manual assessments are labor-intensive, subjective and prone to variability [2]. These limitations have spurred the development of automated image detection systems that provide faster and more accurate evaluations, enhancing both diagnosis and patient care [3]. Automated systems excel in delivering precise CDR measurements, enabling early detection and intervention, which are critical for better clinical outcomes.

The medical image analysis field has made remarkable progress, particularly in segmentation techniques. Earlier methods like thresholding, region growing, and edge detection were some of the first applied to medical imaging [4]. These approaches are known for their computational efficiency and have been widely used for identifying tumors, organs, and other anatomical structures. However, they struggle with variability in image quality, noise, and anatomical differences across patients, often leading to less accurate segmentation results. Recent segmentation techniques, like Active Contour Models, Graph Cuts, and Level Sets,

✉ Sachin Chavan
phd.sachinchavan@gmail.com

[1] Department of Computer Science, SVKM's NMIMS, Mukesh Patel School of Technology Management & Engineering, Shirpur, Maharashtra, India

offer enhanced flexibility and robustness, making them more capable of handling complex shapes and variations in medical images [5]. Despite these advancements, they remain sensitive to initialization parameters, are computationally intensive, and sometimes require manual intervention. The increasing complexity of medical imaging has necessitated a shift toward more advanced solutions.

Machine learning (ML), especially Convolutional Neural Networks (CNNs) [6], has transformed medical image segmentation, achieving cutting-edge results in tasks like optic disc (OD) and optic cup (OC) segmentation for glaucoma diagnosis. Deep learning (DL) models, including Fully Convolutional Networks (FCNs), U-Net and DeepLab, are extensively used because they can automatically learn and extract features from large datasets. This capability enables them to capture both global and local spatial relationships, which are essential for precise segmentation [7]. However, these DL models often require extensive annotated datasets, can be computationally intensive, and may face challenges in adapting to new, unseen data, particularly in medical contexts where annotated datasets are often limited. Several alternative models, such as K-Nearest Neighbours (KNN) [8], which classifies data based on proximity to labeled samples, Naive Bayes [9], a probabilistic classifier that assumes feature independence, XGBoost [10], a gradient boosting framework known for structured data handling, and Deep Belief Networks (DBN) [11], which layer Restricted Boltzmann Machines (RBMs) to learn deep features, have been explored. However, these approaches often struggle to handle the complex spatial patterns inherent in medical images or efficiently scale to larger datasets.

To address the limitations of existing methods, ensemble approaches like AdaBoost and DL models incorporating multi-scale and multi-feature fusion techniques, such as Conditional Random Fields (CRF), have been applied to improve segmentation accuracy [12]. Additionally, unsupervised learning models like Autoencoders [13] have demonstrated efficacy in feature extraction, although they require extensive data. Generative Adversarial Networks (GANs) [14], on the other hand, are particularly beneficial in mitigating data scarcity by generating synthetic medical images, but they demand meticulous tuning and significant computational resources.

Among the array of methods, U-Net [15] emerges as a highly notable model in glaucoma detection, specifically designed for segmenting crucial structures like the OD and OC, which are vital for precise CDR measurement. U-Net's architecture allows for efficient spatial learning, capturing both fine-grained local details and broader contextual information. However, integrating U-Net with pre-trained networks like VGG-16 [16] for feature extraction may impose limitations due to the fixed nature of pre-trained weights, potentially leading to performance bottlenecks, especially in domains with high variance like medical imaging. U-Net also faces challenges with low-contrast images and the complexity of detecting subtle structures amid artifacts such as blood vessels.

The cascaded U-Net architecture, however, offers specific advantages for overcoming these challenges in OD and OC segmentation. It employs a multi-stage approach, progressively refining feature extraction at each stage. This is particularly beneficial for segmenting complex structures, as it enables the model to capture both global and local contextual information more effectively. The cascaded structure also helps in distinguishing between overlapping or adjacent features, such as the OD and OC, by promoting deeper hierarchical learning and minimizing information loss across layers. This method reduces errors in segmenting small structures, such as the optic cup, which are often difficult to delineate in fundus images. Additionally, the cascaded U-Net helps in handling variability in image quality, improving robustness to noise and artifacts like blood vessels, and yielding more accurate segmentation outcomes that are critical for reliable CDR calculation.

Recent progress in segmentation has underscored the necessity for a model that combines the efficiency of traditional methods with the robustness and adaptability of DL techniques, specifically tailored for medical image analysis. The proposed GlaucoAttentNet framework addresses this need by integrating U-Net with attention mechanisms to enhance boundary detection and improve segmentation accuracy. Unlike previous methods, GlaucoAttentNet is designed to detect subtle patterns and variations in fundus images while efficiently handling noise and artifacts. Moreover, it optimizes computational cost, making it a highly accurate and scalable solution for glaucoma detection that can be deployed in resource-constrained environments. This framework also employs advanced pre-processing techniques to improve image quality, ensuring better segmentation of the OD and OC. These enhancements, combined with the cascaded U-Net architecture, enable progressively refined feature extraction, promoting accurate segmentation of complex structures even in challenging images.

1.1 Contribution

The key contributions of the proposed framework include:

- Combined operation of NLM, CLAHE and Adaptive Median filtering modules in sequential pre-processing removes impulse noise adaptively adjusting their parameters based on the local image statistics.
- The encoder within the GlaucoAttent Net progressively creates a hierarchical feature map encompassing both

granular and global image information potentially reducing the model’s sensitivity to noise and bias.
- Introduction of attention mechanisms- Task-specific knowledge attention and spatial attention within GlaucoAttent Net, guided by probability maps highlighting informative regions in fundus images precisely identifying boundaries of the OD and OC, even in the presence of confounding factors like blood vessels.
- We assess our method's performance qualitatively via cross-validation, ROC curve, precision-recall analysis and GlaucoAttent Net's segmentation performance across datasets. Quantitatively, we undertake an ablation study to assess the influence of individual components.

The manuscript is structured as follows: Sect. 2 presents a detailed review of the literature, Sect. 3 describes the proposed methodology and Sect. 4 showcases result as well as conducts a comparative analysis and Sect. 5 concludes with discussions on future directions.

2 Related work

In the realm of glaucoma detection, diverse methodologies have been explored, ranging from DL models and segmentation approaches to innovative hybrid methods. Each avenue contributes unique insights and challenges, as discussed below:

2.1 Domain adaptation models

Domain adaptation is a pivotal aspect of developing robust and generalized models, especially in the context of glaucoma detection. Researchers have explored various approaches to overcome the challenges associated with differences in domain distributions across diverse datasets. Madadi et al. [17] introduced the Glaucoma Domain Adaptation (GDA) model, which utilizes a DL framework incorporating low-rank representation and a weighted source classifier. Despite its effectiveness in aligning domain distributions and facilitating knowledge transfer, the GDA model grapples with challenges related to complexity, substantial computational requirements, and sensitivity to variations in image quality. Another noteworthy approach is the mixup Domain Adaptation (mixDA) method proposed by Yan et al. [18], which combines domain adaptation and domain mixup techniques to enhance glaucoma detection performance. While exhibiting superior results across diverse datasets, this hybrid method is sensitive to parameter settings, necessitating further exploration for comprehensive validation. In a recent study, Pathan et al. [19] introduced the Adaptive Feature Fusion model, providing a novel approach to domain adaptation in glaucoma detection. This innovative model dynamically combines features from both source and target domains, demonstrating improved adaptation capabilities and overall enhanced performance. However, challenges persist, requiring careful optimization of adaptive fusion parameters, highlighting the ongoing complexity of domain adaptation in the field of glaucoma detection.

2.2 Ensemble and attention-based models

Ensemble and attention-based models represent cutting-edge methodologies in the realm of glaucoma detection, where the integration of multiple models or attention mechanisms aims to elevate overall system performance. Resheed et al. [20] introduced an innovative approach by leveraging ensemble learning techniques for glaucoma detection. This involved amalgamating predictions from diverse models, resulting in heightened performance and mitigated overfitting. However, the adoption of ensemble methods introduces challenges related to increased model training complexity and potential computational demands. Liu et al. [21] introduced a deep ensemble network enhanced with an attention mechanism specifically designed for glaucoma screening utilizing stereo optic nerve head images. This model attains superior accuracy through attention-guided utilization of stereo information, albeit with dependencies on image quality and the requirement for specialized expertise and computational resources. On a similar note, Mrad et al. [22] proposed a model that integrates attention mechanisms with ensemble learning, strategically designed to enhance feature selection and boost model generalization. While this approach holds the promise of improved performance through the fusion of attention and ensemble learning, it demands meticulous parameter tuning for both attention mechanisms and ensemble components. Despite the associated challenges, these ensemble and attention-based models offer unique perspectives and advancements in the field of glaucoma detection.

2.3 Segmentation-based approaches

In the realm of glaucoma detection, a segmentation-based strategy is employed to pinpoint and outline specific anatomical structures within medical images, with a particular focus on retinal fundus images of the eye. The fundamental objective of segmentation is to isolate crucial regions or structures linked to glaucomatous pathology, such as the OD and OC. Haider et al. [23] proposed shallow networks like ESS-Net and FBSS-Net, which are designed for OD and OC segmentation in retinal fundus images. These networks exhibit efficient segmentation without the need for extensive pre-processing, although meticulous hyperparameter tuning may be essential. The strengths of these models lie in their accurate segmentation and computational efficiency, yet challenges persist in acquiring substantial training data and addressing potential sensitivity to hyperparameter tuning. Neto et al. [24], on the other hand, employed U-Net models with Inception V3 and Inception ResNet V2 for OD and OC segmentation, showcasing effective glaucoma screening. Challenges include sensitivity to variations in image quality, while the benefits encompass precise segmentation and efficient glaucoma screening, albeit with a reliance on pre-trained models.

2.4 Multi-task learning

Multi-task learning in the field of glaucoma detection involves training a single neural network to simultaneously perform multiple related tasks. A multi-task neural network introduced by Hervella et al. [25] was capable of concurrently handling glaucoma classification and OD/cup segmentation. Their approach demonstrated superior performance in challenging scenarios, eliminating the need for pre-processing or post-processing. However, it introduced different computational requirements compared to baseline multi-task approaches. On a parallel note, Vinayaki and Kalaiselvi [26] introduced the Adaptive Multi-Task Learning model, offering a novel solution to address challenges in glaucoma classification and segmentation. While the optimization of adaptive weights remains an area for further investigation, the model exhibits promising potential to enhance adaptability and overall performance. These contributions signify valuable progress in advancing the capabilities of multi-task learning for glaucoma detection and analysis.

2.5 Deep learning and classification models

These models leverage neural networks and classification techniques to analyze retinal images and identify signs of glaucoma. Thanki [27] introduced a system utilizing DNN with ML classifiers for glaucoma classification, achieving high accuracy and sensitivity. However, the system may be sensitive to dataset characteristics, posing challenges related to potential dataset dependencies and model complexity. Veena et al. [28] employed DL CNN models for OD and OC segmentation, demonstrating high accuracy, but challenges include accurately capturing minute features and high computing requirements. Another approach by Alice et al. [29] utilized a Random Forest classifier for glaucoma classification, incorporating multiple image filters to improve accuracy through comprehensive feature extraction. Challenges for this model involve issues related to model complexity and sensitivity to filter selection.

2.6 Hybrid and innovative approaches

Hybrid and innovative approaches in glaucoma detection integrate diverse methodologies to enhance diagnostic accuracy and overcome challenges. Madhumalini and Devi [30] introduced a fractional gravitational search-based hybrid deep neural network (FGSA-HDNN), achieving superior classification accuracy through unique hybrid features. However, challenges include the demand for extensive datasets and scalability considerations. Agbaloo and Zaccheus [31] explored the Invariant Scattering Convolution Network, achieving high detection correctness while facing challenges related to parameter sensitivity and limited interpretability. Zarza et al. [32] devised a three-stage training methodology for glaucoma detection, showcasing high precision, reliability, and resource efficiency, with challenges and limitations remaining unexplored. D'Souza et al. [33] introduced a transfer learning approach, demonstrating improved performance by utilizing pre-trained models but requiring substantial computational resources. Wu et al. [34] proposed a three-stage training system for glaucoma classification. It starts with robust initial training using data augmentation, followed by fine-tuning with ordinal regression and OD/OC segmentation for better performance. The final stage uses an open evaluation framework for fair assessments and collaboration. This approach yields high precision and reliability but requires careful hyperparameter tuning.

Chen and Pan [35] emphasized interpretability with an Explainable AI Approach balancing transparency and complexity. Shyamalee et al. [36] proposed an Attention-Guided Hybrid Model, optimizing attention and hybrid feature parameters for enhanced feature selection and overall performance. Innovative glaucoma detection methodologies include a Meta-Learning Framework by Shyamalee and Meedeniya [37], enhancing adaptability with challenges in optimizing meta-learning parameters. Zhao et al. [38] explore Capsule Networks, facing parameter optimization challenges but excelling in representation learning. Wassel et al. [39] propose a Vision Transformer-Based Ensemble, overcoming parameter optimization challenges to capture global characteristics for improved classification performance. The Table 1 shows the overview of Sect. 2.

The increasing prevalence of glaucoma necessitates accurate and efficient detection methods. Existing segmentation techniques as discussed above often struggle with generalization due to domain shifts and variability in OD and OC appearances. Challenges such as information overload and insufficient annotated data hinder performance,

Table 1 Overview of the related work

Paper no.	Technique type	Generalization ability	Attention mechanism	Segmentation task	Dataset used
[17]	Deep domain adaptation model	✓	✗	✓	OHTS, ACRIMA, RIM-ONE
[18]	Mixup domain adaptation (mixDA)	✗	✗	✗	REFUGE, LAG, ARCIMA, Drishti-GS
[19]	Ensemble learning with random forests	✗	✗	✓	KMC, RIM, Drishti
[20]	RimNet DL model	✗	✗	✓	Drishti-GS
[21]	Deep ensemble network with attention	✓	✓	✓	Tan Tock Seng Hospital
[22]	Smartphone-based glaucoma screening	✗	✓	✓	DRISHTI-DB, DRIONS-DB, RIAMP
[23]	ESS-Net and FBSS-Net (CNNs)	✓	✓	✓	Drishti-GS, Rim-One-r3, REFUGE
[24]	DL (CNNs)	✗	✗	✓	RIM-ONE r3, DRISHTI-GS, REFUGE
[25]	Multi-adaptive optimization	✗	✗	✓	REFUGE, DRISHTI-GS
[26]	MTRO with R-CNN for diabetic retinopathy	✗	✗	✓	DRIVE
[27]	Dual learning (DL and ML)	✗	✗	✗	HRF, ORIGA
[28]	CNN for OD and OC segmentation	✗	✗	✓	DRISHTI-GS
[29]	CBIR with Random Forest	✗	✗	✓	DB1, DB2, DB3
[30]	DNN with FGSO	✗	✗	✓	RIM-ONE
[31]	Wavelet Image Scattering Network	✗	✗	✗ Focus on classification	RIM-ONE DL
[32]	EfficientNet-based glaucoma detection	✓	✗	✗ Focus on classification	17,070 fundus images
[33]	CNNs and Transformers	✓	✓	✓ U-Net for region segmentation	Rotterdam EyePACS AIROGS
[34]	ML for automated grading	✗	✓	✓ OD and OC segmentation	Largest paired fundus and OCT volumes
[35]	RDR-Net with UDA	✓	✓	✓ OD and OC segmentation	Drishti-GS, RIM-ONE-r3, REFUGE
[36]	DL (segmentation and classification)	✓	✓	✓ OD and OC segmentation	RIM-ONE, ACRIMA, ORIGA, REFUGE, DRISHTI-GS1
[37]	Attention U-Net with CNNs	✓	✓	✓ OD and OC segmentation	RIM-ONE, ACRIMA
[38]	SEDC (Self-ensemble dual-curriculum learning)	✗	✓	✓ OD and OC segmentation	LAG, REFUGE, RIM-ONE
[39]	Vision Transformers (ViTs)	✓	✓	✗ Focus on classification	Merged private datasets

particularly in boundary delineation and class imbalance. Additionally, noise and artifacts in retinal images adversely affect segmentation accuracy, leading to difficulties in precise detection. These limitations underscore the need for improved methodologies that can effectively address these challenges and enhance the reliability of glaucoma diagnosis in clinical practice.

3 Proposed glaucoma detection and severity classification

The proposed Glaucoma detection and severity classification employs a cascaded U-Net with attention mechanisms, named GlaucoAttent Net. Initial steps involve data collection from diverse datasets (ORIGA, REFUGE and G1020) and GFEP using denoising, histogram equalization and adaptive filtering. GlaucoAttent Net integrates encoder, feature fusion and three-stage segmentation paths for accurate OD and OC segmentation. A severity classification module incorporates derived factors, enabling a comprehensive assessment of glaucoma severity. The resulting output integrates probability maps for OD and OC segmentation, providing a three-stage

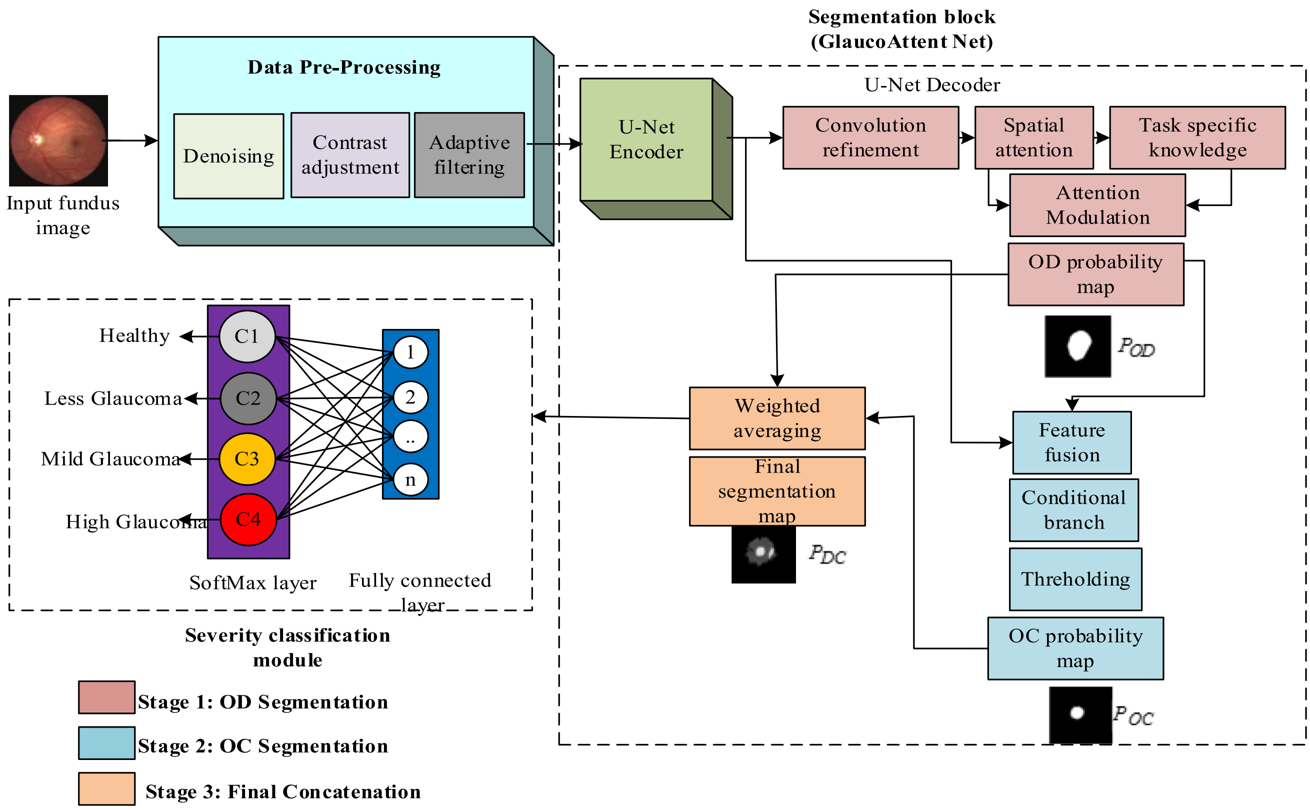


Fig. 1 Proposed methodology overview

representation. The severity classification is achieved through fully connected layers and a SoftMax layer, resulting in a robust glaucoma detection system. Figure 1 provides an overview of the proposed methodology.

3.1 Glaucoma features enhancement pre-processing

Glaucoma Features Enhancement Pre-processing optimizes fundus images in the proposed glaucoma detection framework. It employs NLM denoising, CLAHE and Adaptive Median Filtering to enhance glaucomatous features. It contributes to improved accuracy in diagnosing glaucoma by refining image quality and preserving relevant details.

3.1.1 NLM denoising

The NLM operation begins with the input of a fundus image $x(n)$ containing potential noise arti facts. The algorithm focuses on optimizing the kernel size for the NLM denoising approach, aiming to enhance image quality. The optimizer selects a pixel n within the fundus image and determines an optimized kernel size tailored to the specific characteristics of the image. Subsequently, a search area J is defined around the chosen pixel and linear regression analysis is applied, utilizing pixel intensities from both the current and target kernels. The root mean squared error (RMSE) values for each target kernel are calculated and based on a threshold or criterion, similar pixels are selected for weighted averaging. The weights, determined by Gaussian-weighted Euclidean distance, are employed in a weighted averaging process for noise reduction. The resulting denoised pixel intensity $NKO(x(n))$ reflects the optimized denoising operation. This process is integral in preparing the fundus image for subsequent glaucoma detection algorithms, contributing to improved accuracy in diagnostic tasks. Let us discuss those using mathematical expressions:

In the initial stage, the kernel size α is optimized to enhance the learning performance and convergence speed of the NLM algorithm for fundus image denoising. This optimization occurs through a sequential approach using a stochastic gradient algorithm. The iterative update of the kernel size ($\alpha_{j-1}\prime$) is driven by the prediction error ($r(j)$) and is expressed by the following mathematical formula:

$$\alpha_{j-1}\prime = \alpha_{j-1} + \left(\eta r(j-1)r(j)\alpha_{j-1}^{3}||v(j-1)-v(j)||^{2}\right)\tau_{\alpha_{j-1}}(v(j-1), v(j)) \tag{1}$$

where α_{j-1} represents the current kernel size at iteration $j-1$, $\alpha_{j-1}\prime$ is the updated kernel size, η is a parameter influencing the update, $r(j)$ is the prediction error at iteration j, $\tau_{\alpha_{j-1}}(v(j-1), v(j))$ is the kernel function with the current kernel size.

After optimizing the kernel size, the NLM algorithm is employed to denoise the fundus image. The following steps are involved:

The kernel size and search area are adjusted to accommodate the characteristics of fundus images. This ensures that the NLM algorithm adapts to the specific features of the given image.

The adjustment process involves optimizing the kernel size to balance between preserving fine details and effectively reducing noise. Similarly, the search area is tailored to encompass relevant structures and patterns present in fundus images. The NLM algorithm calculates the denoised intensity $NLM(x(n))$ of a pixel n by computing the weighted average of pixel intensities within the designated search area N.

$$NLM\,(x(n)) = \sum\nolimits_{i\in N} m(n,i).x(i) \tag{2}$$

where $x(i)$ represents the intensity of pixel i in the search area N and $m(n,i)$ is the weight assigned to pixel i for pixel n based on their similarity. The optimization of these parameters ensures that the NLM algorithm effectively captures the unique characteristics of fundus images during the denoising process.

The weight assigned to each pixel is determined by a Gaussian-weighted Euclidean distance ($d(n,i)$) and the overall weights are normalized using a constant ($Y(n)$) is given as:

$$m(n,i) = \frac{1}{Y(n)}.\exp^{-(F.d(n,i))^2} \tag{3}$$

The parameter F adjusts the degree of filtering, influencing the weight distribution.

Following the NLM denoising, a pixel selection method based on linear regression analysis using Root Mean Squared Error (RMSE) values is applied. This step refines the denoising results by excluding dissimilar pixels, enhancing the overall image quality. Parameters such as F and $Y(n)$ are further adjusted according to the specific characteristics of fundus images. Fine-tuning these parameters ensures optimal denoising outcomes tailored to the unique features of the given fundus image.

3.1.2 Contrast limited adaptive histogram equalization (CLAHE)

After optimizing the fundus image using the NLM algorithm with a kernel adapted to its characteristics, the CLAHE process follows, enhancing contrast while considering the image's unique features.

CLAHE divides the image into non-overlapping regions, typically 64 regions for a 512×512 image. Each region is identified by its position (n,i) in the image grid. For each region (n,i), a histogram $h_{n,j}(k)$ is computed. This histogram represents the distribution of pixel intensities in the region, where k ranges from 0 to the maximum intensity value.

The clip limit γ is a crucial parameter in CLAHE, controlling the degree of contrast enhancement. It limits the maximum slope of the cumulative distribution function (CDF) to prevent excessive contrast amplification. The clip limit γ is determined based on a clip factor β and the maximum slope $l_{\max}$ of the histogram. A higher β results in a higher clip limit. It is given as:

$$\gamma = \frac{T_p}{N_g} \times \frac{1 + 100\beta(l_{\max})}{l_{\max}} \tag{4}$$

where T_p is the total number of pixels in the region and N_g is the number of grayscale levels.

The histograms are adjusted to ensure that no pixel intensity exceeds the clip limit γ. This redistribution process involves redistributing pixel counts to limit the maximum number of counts for each grayscale level. The redistribution algorithm iteratively adjusts the histogram until the maximum number of counts for any grayscale level does not exceed the clip limit.

Grayscale mapping involves transforming the histogram to achieve uniform density across grayscale levels. This transformation is achieved through histogram equalization, which redistributes pixel intensities to achieve a more uniform histogram.

The mapping function $f_{n,i}(k)$ for each region (n,i) is derived based on the modified histogram $h_{n,j}(k)$.

For inner regions, each quadrant's pixel intensities are mapped based on the mappings of its four nearest neighboring regions. This ensures that pixel intensities are adjusted based on the local context. For border regions and corner regions, the mapping functions consider the neighborhood structure and adjust pixel intensities accordingly.

Thus, CLAHE dynamically adjusts histograms and performs adaptive histogram equalization to enhance image contrast while preventing over-amplification. The combination of mapping functions ensures that pixel intensities are adjusted based on the local context, resulting in improved image quality.

3.1.3 Adaptive median filtering

After CLAHE, the fundus image undergoes further refinement through Adaptive Median Filtering to enhance noise reduction and maintain image details.

Adaptive Median Filtering is a spatial filtering technique that adjusts the filter size based on the local characteristics of the image. In the context of fundus images, which are often used in medical diagnostics, this process aims to reduce noise while preserving important details for accurate analysis. Here's a step-by-step explanation of what happens in the Adaptive Median Filtering process for fundus images:

The process begins by calculating the color distance map ($\delta_{n,j}$), which represents the differences in color between each pixel and its neighbors. This step is crucial for identifying areas with significant color variations. Two directional maps, the X-map and Y-map, are computed by measuring the color distance between each pixel and its closest right and top neighbors, respectively. These maps highlight the color differences in horizontal and vertical directions.

The X and Y-maps are merged to create a unified color distance map $\left(\delta_{comb}(n,j)\right)$ by selecting the pixel with the higher color distance at each position. This combined map emphasizes pixels with the most substantial color differences.

The chessboard distance is employed to determine the width of the local filtering window in the adaptive median filtering process. It plays a crucial role in assessing the spatial relationships between points in the image and adjusting the filter size accordingly. The chessboard distance ($d_{cb}(F,G)$) is computed as the maximum absolute difference in x and y coordinates between two points F and G.

$$d_{cb}(F,G) = \max\left(\left|x_F - x_G\right|, \left|y_F - y_G\right|\right), \tag{5}$$

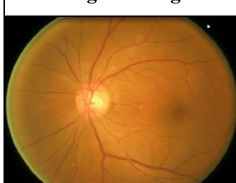
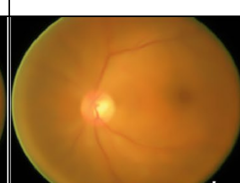
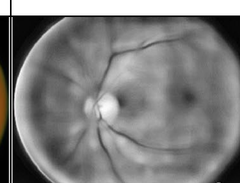
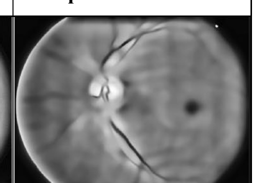
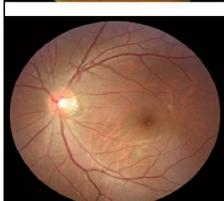
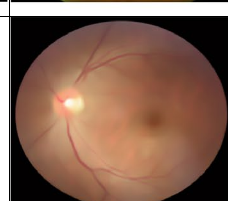
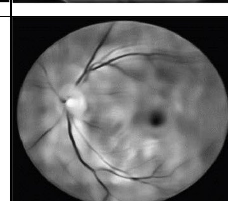
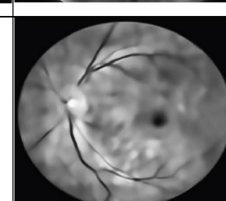
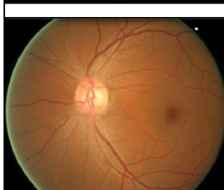
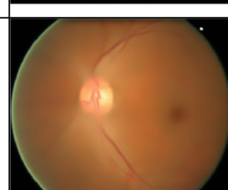
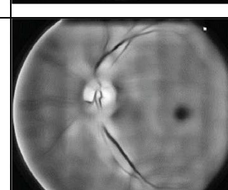
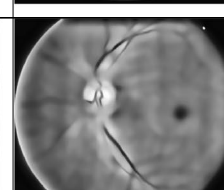
where x_F, y_F are the coordinates of point F and x_G, y_G are the coordinates of point G.

The chessboard distance map ($M_{cb}(n,i)$) is generated by applying the chessboard distance ($d_{cb}(F,G)$) computation across the image. This map characterizes the distance from each background pixel to the nearest object pixel, where objects are pixels with color differences above a specified threshold. This map characterizes the distance from each background pixel to the nearest object pixel, where objects are pixels with color differences above a specified threshold.

Initialize the distance map $\left(M_{cb}\right)$ by setting points of the objects to 0 and those of the background to positive infinity. For each line n of the distance map M_{cb} from top to bottom and for each point (n,i) of the current line, from left to right, the distance $M_{cb}(n,i)$ is set to the minimal value between $M_{cb}(n,i)$,$M_{cb}(n+1,i)+1$,$M_{cb}(n+1,i+1)+1$ and $M_{cb}(n,i+1)+1$. Similar computations are performed for each line n from bottom to top and for each point (n,i) from right to left.

The chessboard distance map provides the half-width of the maximal square included in the background and reaching the objects at least at one point. This essentially calculates the distance from every background point to the closest object. The next step involves determining the width of the local filtering window ($W_f(n,i)$) for each background pixel. It is equal to twice the distance $M_{cb}(n,i)$ plus 1, mathematically expressed as $W_f(n,i) = 2.M_{cb}(n,i) + 1$. For each local region defined by the dynamically determined window size, the median color vector is computed. The color vectors within the region are sorted based on their associated scalar values

Table 2 Dynamic pre-processed fundus image

Original image	NLM	CLAHE	Adaptive median filter
			
			
			

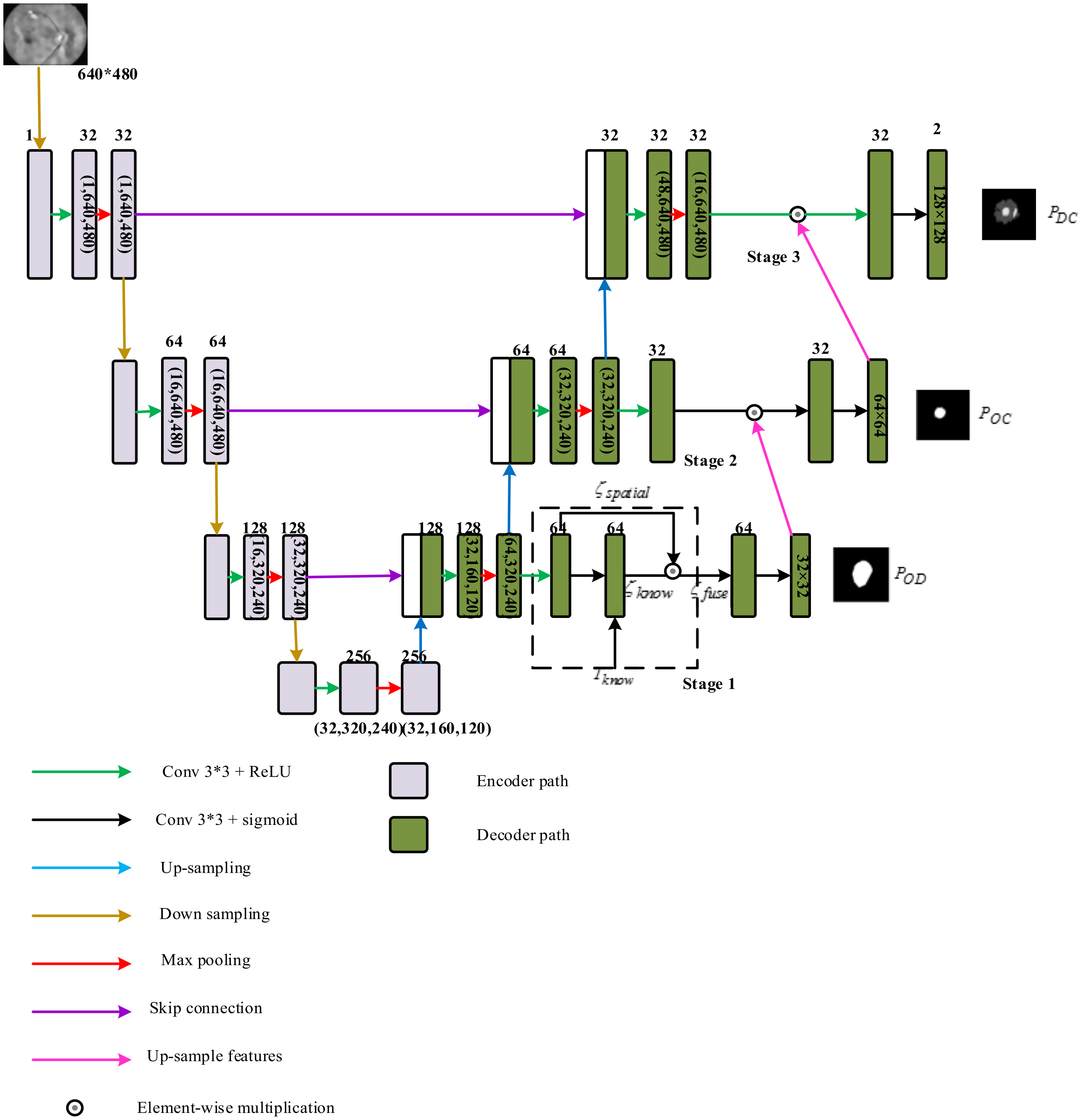


Fig. 2 GlaucoAttent Net architecture

and the median vector is selected. This adaptive median value ensures that the filtering process is sensitive to local variations in color. The adaptive median filtering process is applied to each local region of the retinal image. The final denoised image is obtained by combining the results from all regions. This denoised image exhibits reduced noise while preserving important details, making it suitable for subsequent medical analyses. The pre-processed image in each step is depicted in Table 2.

3.2 GlaucoAttent Net

The grayscale denoised fundus image ($x'(n)$) is the initial input to the encoder path of the U-net based GlaucoAttent Net. The architecture of GlaucoAttent Net is given in Fig. 2. Initially, pixel intensities in the input image are normalized to ensure uniformity and compatibility with the neural network model. The pixel values are scaled linearly to a range between 0 and 1.The formula for min–max normalization is given by:

$$X_{norm} = \frac{X - \min(X)}{\max(X) - \min(X)} \tag{6}$$

where X represents the original pixel values and $\min(X)$ and $\max(X)$ are the minimum and maximum pixel values in the image, respectively. Then the normalized fundus image enters the encoder path.

3.2.1 Encoder path

In the initial layers of the encoder path, the convolutional operations act as filters to detect local patterns and variations in pixel intensities. The convolutional filters have small receptive fields, enabling them to capture local details in the input image such as, a convolutional filter detect edges by responding to intensity changes in adjacent pixels. The mathematical operation for convolution at a given layer l can be expressed as:

$$Z_l = W_l * x'(n) + \beta_l \tag{7}$$

where Z_l represents the feature map, W_l is the convolutional filter, $x'(n)$ is the input image and β_l is the bias term.

Batch normalization normalizes the activations, including the intensity variations captured by convolution. It ensures that the features have zero mean and unit variance, facilitating stable training. The scaling factor (Φ_l) and shift term (β_l) in batch normalization allow the network to adapt the intensity levels. It is given as

$$B_l = \frac{Z_l - \mu_l}{\sqrt{\sigma_l^2 + \in}}.\Phi_l + \beta_l \tag{8}$$

where Z_l's standard deviation and mean are given as σ_l and μ_l.

The ReLU activation function (A_l) introduces non-linearity by setting negative values to zero. This non-linearity is crucial for capturing variations in pixel intensities. If the intensity is below a given threshold, ReLU will output zero, effectively capturing regions of low intensity. It can be expressed as:

$$A_l = \max(0, B_l) \tag{9}$$

Then Max pooling is applied to reduce spatial dimensions while retaining significant features. It identifies regions of maximum intensity, helping to preserve crucial information about intensity variations at a lower resolution.

$$P_l = \max(A_l) \tag{10}$$

Thus

$$E_l = P_l(A_l(B_l(W_l(E_{l-1})))) \tag{11}$$

where E_{l-1} is the output from the previous stage.

After encoder path, the Feature Fusion Block in the U-Net architecture is responsible for combining high-level features from the final encoder layer with features from different decoder stages. This block plays a crucial role in preserving both low-level details (intensity gradients and local structures) and high-level semantic information (texture patterns, boundary information, OD structure and global relationships), ensuring that the decoder can make use of relevant information from various hierarchical levels like low level and high level information. The feature fusion process enhances the network's ability to reconstruct detailed structures during the decoding phase.

Let's denote the features from the final encoder layer as E_{final} and features from different decoder stages as D_s for each stage s. The Feature Fusion Block operates as follows:

$$F_{fuse} = concat(E_{final}, D_1, D_2, \ldots D_n) \tag{12}$$

Here, *concat* represents the operation of concatenating the feature maps along the stage dimension. This results in a combined feature map F_{fuse} that contains information from the final encoder layer as well as multiple decoder stages.

The concatenation operation can be expressed as:

$$F_{fuse}[u,v,k] = \begin{cases} E_{final}[u,v,k] & if\ 1 \le k \le C_{final} \\ D_1[u,v,k - C_{final}] & if\ C_{final} < k < C_{final} + C_1 \\ D_2[u,v,k - (C_{final} + C_1)] & if\ C_{final} + C_1 < k \le C_{final} + C_1 + C_2 \\ \ldots. & \ldots. \\ D_n[u,v,k - \sum_{m=1}^{n-1} C_m] & if\ \sum_{m=1}^{n-1} C_m < k \le \sum_{m=1}^{n} C_m \end{cases} \tag{13}$$

where C_{final} is the number of stages in the final encoder layer, $C_1, C_2, ..., C_n$ are the number of stages in each decoder stage, u, v are spatial indices, k is the stage index in the concatenated feature map.

This process allows the network to have access to both detailed features from the early stages of the encoding process and high-level semantic features from the final encoder layer, facilitating more accurate and context-aware segmentation during the decoding phase.

3.2.2 Decoder path

The decoder path in our proposed methodology, GlaucoAttent Net, plays a pivotal role in refining the OD and OC segmentation process. This component acts as the counterpart to the encoder, reconstructing high-resolution feature maps from the abstract representations generated by the encoder. Through a series of upsampling and concatenation operations, the decoder path allows the model to precisely localize and accurately delineate the boundaries of the OD and OC. This is well explained in the following three stages producing three stage outputs.

3.2.2.1 Stage 1: Cascaded U-Net with attention mechanisms—OD segmentation In the first stage of the proposed Cascaded U-Net with Attention Mechanisms for glaucoma detection, the focus is on accurately segmenting the OD from fundus images. Building on the features extracted in the encoder path and refined through the feature fusion block, the algorithm employs attention mechanisms for precise segmentation.

The journey begins with the output from the Feature Fusion Block, denoted as F_{fuse}. This feature map integrates information from both the final encoder layer capturing high-level semantic details and features from the first decoder stage, consolidating both low and high-level features.

The concatenated feature map $\left(F_{fuse}\right)$ undergoes convolutional refinement (conv5) with a specific set of learnable parameters (L_{conv5}). Mathematically, the operation is defined as:

$$F_{conv5} = Conv(F_{fuse}, L_{conv5}) \tag{14}$$

Simultaneously, a separate branch focuses on spatial attention to highlight regions in the image more likely to belong to the OD. A convolutional layer followed by a sigmoid activation produces a spatial attention map ($\zeta_{spatial}$).

$$\zeta_{spatial} = \sigma(Conv(F_{fuse}, L_{attent5})) \tag{15}$$

In addition to spatial attention, task-specific knowledge about the OD, such as its average size and shape, is incorporated. Let's denote the pre-defined tensor representing task-specific knowledge as T_{know}. This tensor encapsulates information about the expected characteristics of the OD, providing guidance to the model. The knowledge-based attention map (ζ_{know}) derived through a dedicated convolutional operation. This involves convolving the features F_{conv5} with learnable parameters specific to the attention mechanism (L_{know}).

$$\zeta_{know} = \sigma(Conv(F_{conv5}, L_{know})) \tag{16}$$

Here, σ is the sigmoid activation function, ensuring that the resulting attention weights are in the range [0, 1].

The obtained attention map ζ_{know} represents how much attention the model should allocate to each spatial location based on the incorporated knowledge. This map is then combined with the spatial attention map ($\zeta_{spatial}$) derived in the previous step using element-wise multiplication.

$$\zeta_{fuse} = \zeta_{spatial} \cdot \zeta_{\mathrm{know}} \tag{17}$$

Element-wise multiplication enables the model to account for the spatial context (from $\zeta_{spatial}$) and the task-specific knowledge (from ζ_{know}) when determining the final attention. Thus ζ_{fuse} forms the combined OD attention map.

The features from F_{conv5} are modulated by the combined attention map (ζ_{fuse}). Subsequently, the modulated features undergo up-sampling and convolutional operations (L_{conv6}) to obtain a refined feature map (F_{conv6}).

$$F_{conv6} = UpConv(Mod(F_{conv5}, \zeta_{fuse}), L_{conv6}) \tag{18}$$

Finally, a sigmoid activation layer is applied to F_{conv6} so that the probability map (P_{OD}) can be generated for the OD segmentation:

$$P_{OD} = \sigma(F_{conv6}) \tag{19}$$

The stage1 output is derived using Eq. (20)

3.2.2.2 Stage 2: OC segmentation within predicted OD region Stage 2 of the segmentation model focuses on segmenting the OC within the predicted OD region. This stage involves several key steps, including feature fusion with attention-weighted features, a conditional branch for cup presence probability, and the main decoder branch for OC segmentation.

In Feature Fusion with Attention-weighted Features step, features from different decoder stages (D_s) are concatenated to form F_{fuse}. The fusion process facilitates the integration of information from different scales and resolutions. The combined attention map for the OD (ζ_{fuse}) is then applied to weight the features in F_{fuse}, generating weighted features M_f. This can be expressed as:

$$M_f = \zeta_{fuse} \cdot F_{fuse} \tag{20}$$

Here, • represents element-wise multiplication and M_f now emphasizes regions in F_{fuse} that are more relevant to the predicted OD.

A separate branch (*cond_br*) uses convolutional layers to predict a probability map (OC_p) indicating the likelihood of each pixel belonging to the optic cup.

$$OC_p = Conv(M_f) \tag{21}$$

Here, *Conv* represents the operations performed by the convolutional layers in the conditional branch. These layers capture spatial patterns and relationships within the attention-weighted features, refining the information to make predictions about the presence of the optic cup. The output of these layers is a feature map, and a sigmoid activation function is commonly applied to this feature map to obtain pixel-wise probabilities. The sigmoid activation function squashes the values between 0 and 1, representing the probability of each pixel belonging to the optic cup.

$$OC_p = \sigma(Conv(M_f)) \tag{22}$$

The resulting OC_p map represents the likelihood of cup presence at each pixel in the predicted OD region. This probability map is crucial for generating the final segmentation of the OC in the given image.

Application of threshold to the cup presence map (OC_p) obtained from the conditional branch is done. This thresholding operation separates pixels with high cup probability from those with low cup probability.

$$OC_{mask} = \begin{cases} 1, & if\ OC_p \geq threshold \\ 0, & otherwise \end{cases} \tag{23}$$

Here, OC_{mask} represents the binary mask indicating regions of high cup probability based on the threshold value.

Now, the OC segmentation process unfolds in the main decoder branch. Similar to the OD segmentation pathway, M_f are up-sampled and further processed through convolutional layers in the main decoder branch. The convolutional layers refine the features to extract higher-level representations relevant to cup segmentation. These refined features are given as D_{out}. The final segmentation probability mask involves element-wise multiplication (•) between the main decoder output and the binary mask.

$$P_{OC} = D_{out} \cdot OC_{mask} \tag{24}$$

Thus, producing stage 2 output by Eq. (25)

Table 3 GlaucoAttent Net layer details

Layer type	layer size	Filter size	Stride	Number of filters
Input image	(256, 256, 1)	–	–	–
Conv2D (encoder)	(256, 256, 32)	(3, 3)	1	32
Conv2D (encoder)	(256, 256, 32)	(3, 3)	1	32
MaxPooling2D (encoder)	(128, 128, 32)	(2, 2)	2	–
Conv2D (encoder)	(128, 128, 64)	(3, 3)	1	64
Conv2D (encoder)	(128, 128, 64)	(3, 3)	1	64
MaxPooling2D (encoder)	(64, 64, 64)	(2, 2)	2	–
Conv2D (encoder)	(64, 64, 128)	(3, 3)	1	128
Conv2D (encoder)	(64, 64, 128)	(3, 3)	1	128
MaxPooling2D (encoder)	(32, 32, 128)	(2, 2)	2	–
Conv2D (encoder)	(32, 32, 256)	(3, 3)	1	256
Conv2D (encoder)	(32, 32, 256)	(3, 3)	1	256
Feature fusion block	–	–	–	-
Conv2D (decoder)	(32, 32, 32)	(3, 3)	1	32
Conv2D (decoder)	(32, 32, 32)	(3, 3)	1	32
Conv2D (decoder)	(32, 32, 1)	(3, 3)	1	1
Concatenation	(32, 32, 129)	–	–	–
Conv2D (decoder)	(32, 32, 64)	(3, 3)	1	64
Conv2D (decoder)	(32, 32, 1)	(3, 3)	1	1
Concatenation	(32, 32, 130)	–	–	–
Conv2D (decoder)	(32, 32, 64)	(3, 3)	1	64
Conv2D (decoder)	(32, 32, 1)	(3, 3)	1	1
Concatenation	(32, 32, 131)	–	–	–

3.2.2.3 Stage 3: Final concatenated output (P_{DC}): The process begins by running the U-Net model on the fundus image to obtain two distinct probability maps—P_{OD} and P_{OC}. These probability maps capture the likelihood of each pixel belonging to the OD and the presence of OC within the predicted OD region, respectively.

Next, the original probability maps are combined by stacking them along the stage dimension. This operation results in a multi-stage image where Stage 1 represents the P_{OD} probability map, Stage 2 represents the P_{OC} probability map and a Stage 3 is introduced to capture combined information. Thus, the final output image is a three-stage representation, with each stage encapsulating crucial information about the OD and OC. Stages 1 and 2 delineate the probability of OD segmentation and cup presence, respectively. Stage 3, offers a tailored combination of information from the previous stages, further refined through normalization. The stage 3 is represented as:

$$P_{CD} = \frac{(P_{OD}.w_{OD}) + (P_{OC}.w_{OC})}{w_{OD} + w_{OC}} \tag{25}$$

where w_{OD} and w_{OC} are the weight factors that control the influence of each probability map. This equation represents a weighted average of the two probability maps, where the weights are determined by w_{OD} and w_{OC}. The normalized result in stage 3 provides a tailored combination of information from both OD and OC probability maps. Adjusting the weights allows for fine-tuning the contribution of each map to the final concatenated output. The Table 3 shows the GlaucoAttent Net layer details.

3.3 Severity classification module

The severity classification module in the glaucoma detection system plays a crucial part in evaluating the severity condition by analysing factors derived from the segmentation results including Optical CDR, OD and OC Areas, Diameters, Location, Contour Features and Rim Thinning. These factors collectively contribute to a comprehensive understanding of glaucoma severity, capturing both structural and spatial characteristics of the OD and OC. Table 4 give the factors considered during detection and classification.

The initial step involves the formation of a feature vector (X), where each element represents one of the derived factors. This feature vector is essential for consolidating the relevant information into a format suitable for further processing. Mathematically, the feature vector can be denoted as $X = [X_1, X_{2,}..., X_6]$, where we have only 6 derived factors.

Subsequently, the feature vector X undergoes transformation through a series of FCN layers. At each FCN layer (FC_i), the input vector undergoes multiplication with a weight matrix, followed by a bias vector addition. Subsequently, the resultant output is passed through an activation function, commonly ReLU, introducing non-linearity to the model. This process is iterated through the sequence of fully connected layers until reaching the final layer, which generates the output vector Out_{last}.

The final stage involves the SoftMax layer, where the output vector Out_{last} is processed through the SoftMax function. The SoftMax function computes the probability distribution over different glaucoma severity classes. Each element in the SoftMax output indicates the probability of the input belonging to its corresponding severity class. Mathematically, the SoftMax function is expressed as:

$$Soft\max(Out_{last})_i = \frac{e^{Out_{last,i}}}{\sum_{j=1}^{C} e^{Out_{last,j}}} \tag{26}$$

Here, C represents the total number of severity classes. Each element in the SoftMax output indicates the probability of the input belonging to its corresponding severity class.

4 Experimental details

In this section, we present a comprehensive analysis of three datasets, an experimental setup, evaluation metrics, results and analysis sections shedding light on performance of our proposed glaucoma detection and severity classification.

4.1 Experimental set up and its details:

4.1.1 Dataset description

4.1.1.1 Dataset 1 The ORIGA dataset [40] is a collection of retinal fundus images specifically designed to support research in glaucoma. This dataset originates from the Singapore Malay Eye Study (SiMES) conducted by the Singapore Eye Research Institute (SERI). ORIGA comprises a total of 650 high-quality retinal images meticulously marked by experts to ensure accurate labeling. The dataset offers a balanced distribution of images, with 168 images representing cases diagnosed with glaucoma and the remaining 482 images belonging to non-glaucoma cases. This balanced representation allows researchers to effectively analyze both healthy and diseased retinas.

4.1.1.2 Dataset 2 The REFUGE dataset [41] is a game-changer for researchers and practitioners working on automated glaucoma detection and OD/cup segmentation. Introduced by Orlando et al., REFUGE stands out as the largest existing dataset specifically designed for this purpose, boast-

Table 4 Considered factors

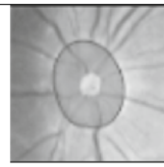
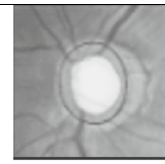
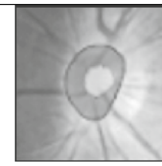
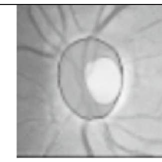
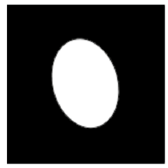
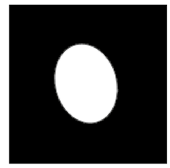
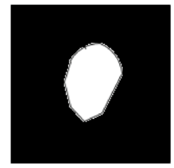
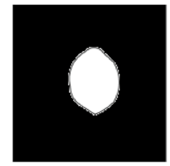
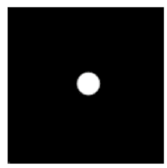
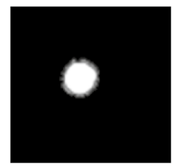
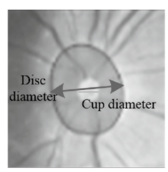
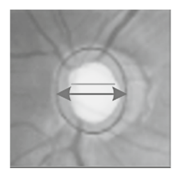
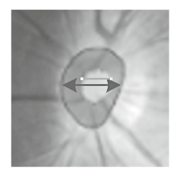
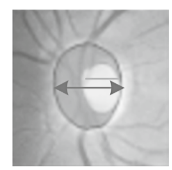
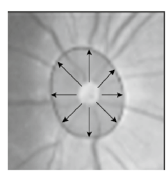
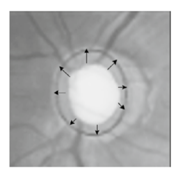
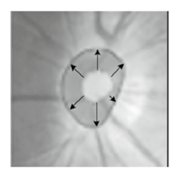
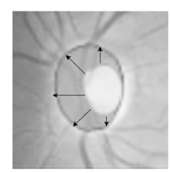
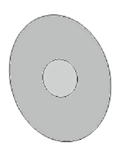
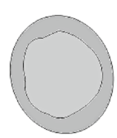
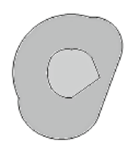
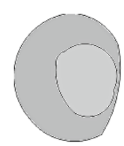
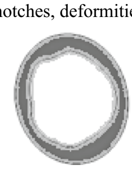
Factors	Healthy Eyes	Highly Glaucomatous Eyes	Less Glaucomatous Eyes	Moderate Glaucomatous Eyes
CDR	 Normal and balanced $0.3 \leq CDR \leq 0.6$	 Markedly increased $CDR > 0.6$	 Slightly increased $0.2 \leq CDR < 0.3$	 Moderately increased $0.3 < CDR \leq 0.4$
Disc and cup Areas				
	 Moderate, proportional	 Noticeably enlarged	 Mildly enlarged	 Moderately enlarged
Diameter	 Normal	 Increased, especially in the cup	 Slightly increased, especially cup	 Moderately increased
Location	 Even distribution	 Abnormalities concentrated	 Some abnormalities present	 Noticeable abnormalities
Contour Features	 Smooth and regular	 Irregularities, notches, deformities	 Mild irregularities	 Moderate irregularities
Rim Thinning	 Adequate and uniform	 Significant thinning	 Mild thinning	 Moderate thinning

Table 5 Dataset Description

Dataset	Dataset 1	Dataset 2	Dataset 3
Name	ORIGA	REFUGE	G1020
Resolution	High-resolution (color) 3072 × 2048	High-resolution 2124 × 2056	High-resolution (color) 1944 × 2108
Camera	Acquired through SiMES by SERI	Not specified	Not specified
Number of Samples	650	1200	1020
Annotations	168 glaucomatous and 482 non-glaucomatous image	Ground Truth Segmentation (OD and OC), Clinical Labels (Glaucoma diagnosis)	Glaucoma Diagnosis (Healthy or glaucoma), Segmentation (OD and OC), Quantitative Measurements (Vertical CDR, Size of neuroretinal rim), Spatial Information (Bounding box location for OD)
Year of Release	2010	2018	2020

Table 6 Dataset Partitioning

Dataset	Training Samples	Validation Samples	Testing Samples
ORIGA	390 (60%)	130 (20%)	130 (20%)
REFUGE	720 (60%)	240 (20%)	240 (20%)
G1020	612 (60%)	204 (20%)	204 (20%)

Table 7 Parameter settings

Parameters	Trained on
Batch size	16
Training Epochs	50
Loss function	Binary Cross-Entropy
Optimizer	Adam
Learning rate	0.001

ing a collection of 1200 meticulously annotated retinal fundus images. These images come with valuable ground truth information, including precise segmentation outlines for the OD and OC, as well as clinical labels confirming glaucoma diagnosis or a healthy state. This comprehensive dataset empowers researchers to develop and rigorously evaluate automated diagnostic methods, ultimately leading to significant advancements in glaucoma assessment techniques.

4.1.1.3 Dataset 3 The G1020 dataset stands as a landmark resource in the fight against glaucoma, a vision-threatening disease. Introduced by Bajwa et al. [42], it has become the benchmark dataset for evaluating algorithms designed for computer-aided glaucoma detection. This valuable dataset comprises 1020 meticulously curated, high-resolution color fundus images, reflecting real-world ophthalmology practices. Each image includes comprehensive ground truth annotations, offering vital insights for glaucoma diagnosis and OD analysis. These annotations encompass clear labels indicating glaucoma presence or absence, precise segmentation of OD and OC areas and essential quantitative measurements such as vertical CDR and neuroretinal rim size. Furthermore, the dataset furnishes spatial information via bounding box coordinates for the OD. The dataset details are given in Table 5.

In order to ensure the model's generalization capability and prevent data leakage, we rigorously separated the datasets into distinct training, validation, and testing subsets. Crucially, images from the same patient are excluded from appearing in more than one subset. This precaution eliminates the risk of the model memorizing specific features from an individual patient and ensures that the model's performance is evaluated on unseen data. The number of images in each subset is as given in Table 6:

4.1.2 Experimental set up

In this research, Careful consideration is given to dataset selection to ensure a broad scope of glaucoma cases. Preprocessing emerged as a crucial step to enhance image quality. NLM Denoising, a pivotal technique, is applied with optimized kernel sizes using a sequential approach and a stochastic gradient algorithm. Fine-tuning parameters like α (kernel size), η (update parameter), and τ (kernel function) are imperative for achieving optimal denoising performance. Additionally, CLAHE and Adaptive Median Filtering are employed to further refine image contrast and mitigate noise.

The GlaucoAttent Net, tailored for accurate segmentation, boasted an input size of 256 × 256x1, specifically designed for grayscale fundus images. Comprising an encoder, feature fusion block and decoder, the network integrated attention mechanisms, encompassing both spatial attention and knowledge-based attention. Training parameters are carefully chosen, utilizing a binary cross-entropy loss function for segmentation tasks, the Adam optimizer, a learning rate of 0.001, a batch size of 16 and a training duration spanning

Table 8 Memory consumption details

Component	Memory consumption (MB)	Notes
Input images (pre-processed)	6	Memory for storing input images (e.g., 512 × 512x3 images)
NLM denoising	64	Memory for patch similarity matrices (per image)
CLAHE processing	1	Minimal additional memory usage
Adaptive median filtering	2	Slight increase for storing filtered images
Encoder feature maps	32	Memory for storing feature maps at various levels
Feature fusion block	16	Memory for storing fused features
Decoder feature maps	64	Memory for intermediate feature maps during upsampling
Attention Mechanisms	16	Memory for attention weights and maps
Model parameters	40	Memory for model weights and biases
Optimizer states	8	Memory for storing optimizer states (e.g., Adam)
Total memory usage	233	Estimated total memory consumption

50 epochs. The severity classification module, leveraging factors like Optical CDR and diverse contour features, employed a feature vector of dimension 6. This module comprised FCN layers with ReLU activation and a softmax layer for severity classification.

The experimental setup is executed on a high-performance machine configured with an NVIDIA GeForce RTX 2080 Ti GPU, which significantly accelerated the training and evaluation processes due to its robust parallel processing capabilities. The system operated on Windows 10, a widely used operating system that provides strong support for DL frameworks and hardware compatibility. The software environment is meticulously prepared using Python 3.12, the latest version, which provided the necessary features and libraries for ML algorithms efficiently.

For the DL model development, TensorFlow 2.15.0 is utilized, offering advanced functionalities such as eager execution and a flexible architecture conducive to building complex neural networks. Keras 2.4.3, integrated with TensorFlow is employed for its user-friendly interface, which simplified the process of defining and training the GlaucoAttentNet architecture. To ensure optimal performance on the GPU, the setup includes CUDA 11.2, which enables GPU acceleration for tensor computations, significantly enhancing the speed of model training. The experimental framework also includes key libraries such as NumPy for numerical operations, OpenCV for image processing and Matplotlib for visualizing results. Furthermore, all necessary dependencies are managed using pip to maintain a consistent environment, facilitating reproducibility of the experiments. The parameter settings are given in Table 7.

The proposed GlaucoAttentNet methodology involves significant memory usage across various components as shown in Table 8. Key contributors include NLM denoising, which consumes around 64 MB for patch similarity matrices, and the decoder feature maps, adding approximately 64 MB for intermediate results. The cascaded U-Net architecture and attention mechanisms also contribute, with the encoder and feature fusion block requiring around 32 MB and 16 MB, respectively. Model parameters and optimizer states add 48 MB. Overall, the total estimated memory consumption is 233 MB.

4.1.3 Evaluation metrics:

These metrics offer a thorough evaluation of the model's performance, encompassing various facets of its precision, accuracy and capacity to manage positive and negative instances effectively.

• Accuracy:

Accuracy assesses the overall correctness of the model by considering both true positives (correctly predicted positive instances) and true negatives (correctly predicted negative instances).

$$Acc = \frac{(Pos_T + Neg_T)}{(Pos_T + Neg_T + Pos_F + Neg_F)} \tag{27}$$

Precision ($\Pr e$):

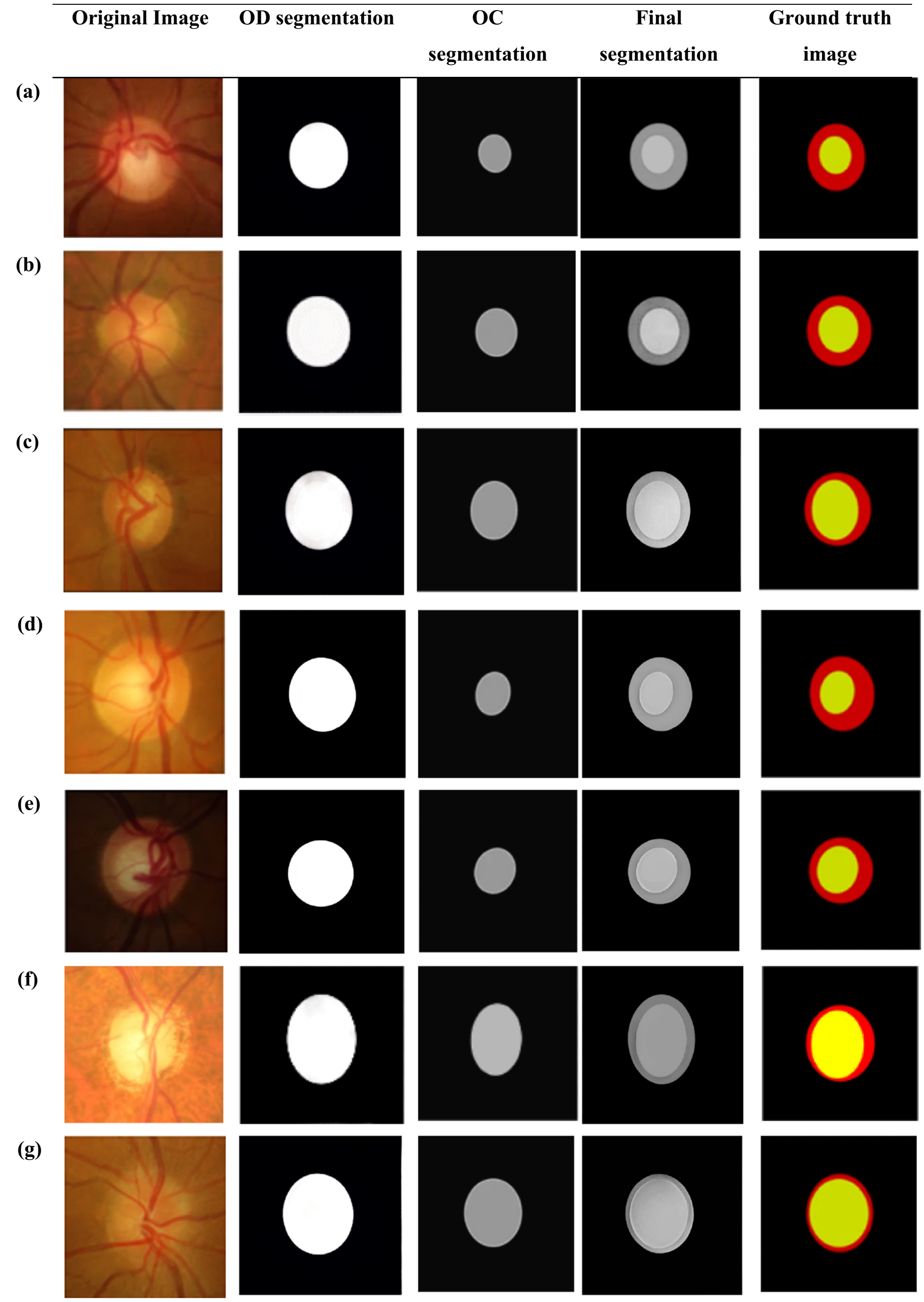
Precision evaluates the correctness of positive predictions generated by the model. It is calculated as the proportion of true positives among all instances predicted as positive.

$$\Pr e = \frac{Pos_T}{(pos_T + Pos_F)} \tag{28}$$

• F-Score

The F-Score represents the harmonic mean of precision and recall, offering a balanced assessment of these two metrics. It proves beneficial in scenarios where class distribution is skewed.

Table 9 Segmentation result



$$F\_Score = \frac{2 * (pre * rec)}{(pre + rec)} \tag{29}$$

Sensitivity (true positive rate (rec)):

Recall evaluates the model's capability to identify all positive instances accurately. It is calculated as the ratio of true positives to the total actual positive instances.

$$rec = \frac{Pos_T}{(Pos_T + Neg_F)} \tag{30}$$

Specificity (true negative rate ($spec$)):

Specificity evaluates the model's accuracy in identifying negative instances correctly. It is computed as the ratio of true negatives to the total actual negative instances. Depending on the objectives of the glaucoma detection task, various metrics may take precedence.

$$Spec = \frac{Neg_T}{(Neg_T + Pos_F)} \tag{31}$$

4.2 Result and analysis

The results and analysis section serves as the crucible where the culmination of research efforts is unveiled and dissected. In this pivotal segment, we meticulously dissect the outcomes of our study, laying bare the empirical evidence that emerged from experimental endeavours.

4.2.1 Qualitative analysis

The qualitative analysis of the proposed segmentation method is given below.

The qualitative analysis of segmentation results as shown in Table 9 compares OD, OC and final segmentation outcomes from the ORIGA, REFUGE, and G1020 datasets against ground truth annotations. Images (a), (b), and (c) are from ORIGA, (d) and (e) from REFUGE, and (f) and (g) from G1020. This visual comparison highlights the model's accuracy in delineating OD and OC regions, showcasing strong agreement with expert annotations. The segmentation results demonstrate robustness across datasets, with minimal deviations from the ground truth. This analysis complements quantitative evaluation, validating the model’s effectiveness in capturing glaucoma-related retinal features.

4.2.2 Cross-validation

The glaucoma detection framework proposed employs a robust fivefold cross-validation process to ensure comprehensive model evaluation. This process is applied to the combined dataset, which integrates the ORIGA, G1020, and REFUGE datasets. The combined dataset includes all three to ensure diversity and representation for robust training. The data is split into training (60%), validation (20%), and testing (20%) sets, and the cross-validation is conducted only on the training set to optimize model performance. In fivefold cross-validation, the training set is divided into five equal subsets. During each iteration, one subset is used as the validation set while the remaining four subsets are used for training. This ensures that each subset serves as the validation set

Table 10 fivefold cross-validation performance metrics

Metric	I1	I2	I3	I4	I5	Average	Std. Dev
Accuracy	0.92	0.91	0.93	0.90	0.92	0.916	0.012
Precision	0.89	0.91	0.92	0.88	0.90	0.900	0.015
Recall	0.94	0.92	0.95	0.91	0.93	0.930	0.016
F1-Score	0.91	0.91	0.93	0.89	0.91	0.910	0.013

Table 11 10-fold cross-validation performance metrics

Metric	I1	I2	I3	I4	I5	I6	I7	I8	I9	I10	Average	Std. Dev
Accuracy	0.92	0.91	0.93	0.90	0.92	0.91	0.93	0.90	0.92	0.94	0.916	0.012
Precision	0.89	0.91	0.92	0.88	0.90	0.90	0.91	0.88	0.89	0.92	0.900	0.015
Recall	0.94	0.92	0.95	0.91	0.93	0.94	0.95	0.92	0.94	0.96	0.930	0.016
F1-Score	0.91	0.91	0.93	0.89	0.91	0.92	0.93	0.89	0.90	0.94	0.910	0.013

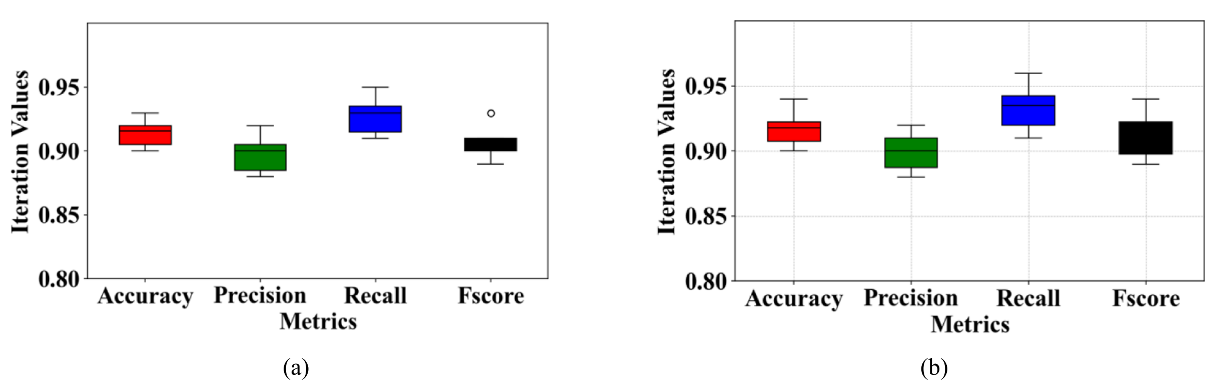


Fig. 3 The box plot for **a** fivefold and **b** tenfold cross validation

exactly once, with the process repeated over five iterations. After each iteration, performance metrics are calculated on the validation set to provide an evaluation of the model's performance across different data configurations. The fivefold cross-validation results are summarized in Table 10 below.

These performance metrics are calculated for each iteration, performance metrics are computed and then averaged to offer a thorough assessment of the model's effectiveness. The inclusion of standard deviation helps gauge the variability in performance across various folds.

Following each iteration, model hyperparameters can be fine-tuned based on the cross-validation results, optimizing its ability to detect glaucoma features effectively. The final model is then trained on the entire dataset using the optimized hyperparameters for comprehensive evaluation and potential deployment.

A tenfold cross-validation process is conducted independently on the test set of the combined dataset. The test set is divided into ten equal subsets, with one subset used as the validation set in each iteration while the remaining nine subsets are utilized for training the model. Each subset is used as the validation set exactly once, and performance metrics are computed for each iteration on the respective validation subset. This approach ensures robust evaluation and enhances the model's ability to generalize to unseen test data. The tenfold cross-validation results are summarized in Table 11 below.

This cross-validation approach ensures that the proposed glaucoma detection framework is robust, reliable and capable of generalizing well to diverse datasets, mitigating overfitting and providing a thorough assessment of its diagnostic accuracy. The box plot for cross validation of 5 and 10 folds is given in Fig. 3.

4.2.3 Roc curve analysis

In the ROC analysis, the discrimination threshold of the model is varied to assess each dataset's performance in classifying positive and negative instances. *rec* and *spec* are calculated across different threshold values. Sensitivity measures the model's accuracy in detecting positive instances, while specificity evaluates its precision in classifying negative instances. Our proposed methodology achieved an impressive Area Under the Curve (AUC) of 0.99 for the ORIGA dataset, indicating excellent discriminatory ability and balance between true positive and false positive rates. The ROC curve visually depicts the model's performance across various sensitivity and specificity values.

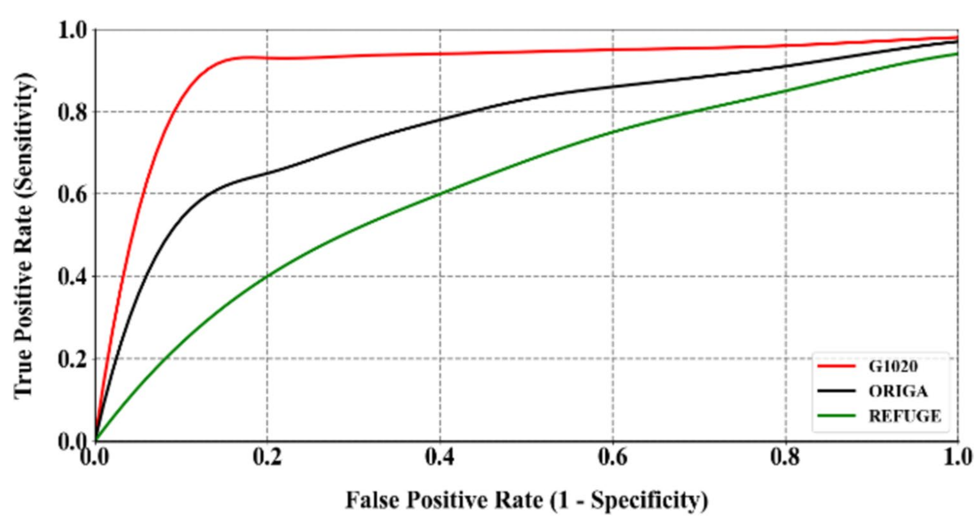
Similarly, for the REFUGE dataset, our proposed methodology demonstrated strong performance with an AUC of 0.98. The ROC curve for REFUGE provides a comprehensive insight into the model's behavior under different threshold configurations, facilitating the determination of an optimal operating point tailored to the application's specific needs. In the case of the G1020 dataset, our methodology exhibited a robust AUC of 0.93. The ROC curve for G1020 allows for a nuanced examination of the model's performance characteristics, offering insights into its ability to generalize and adapt to diverse datasets.

Overall, ROC analysis for our proposed methodology shown in Fig. 4 provides a comprehensive understanding of its classification performance across multiple datasets.

4.2.4 Comparative analysis

The bar graph presented in Fig. 5 provides a comprehensive visualization of the precision, recall, and F-measure scores for various glaucoma detection methods, including

Fig. 4 ROC graph



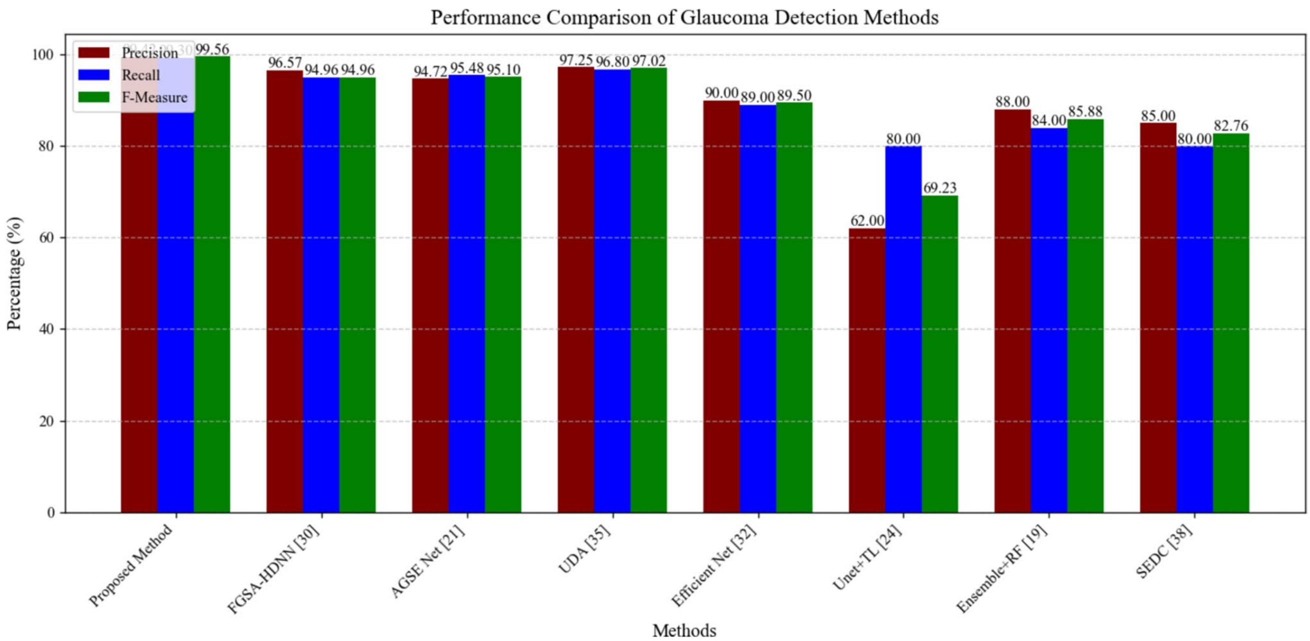


Fig. 5 Precision–recall analysis graph

FGSA-HDNN [30], AGSE Net [21], UDA [35], Efficient Net [32], Unet + TL [24], Ensemble + RF [19], SEDC [38], and the proposed method. Each of these metrics plays a crucial role in assessing the effectiveness of the methods in accurately detecting glaucoma. In terms of precision, the graph illustrates that the proposed method outperforms all other methods, achieving an exceptional score of 99.42%. This indicates that it accurately identifies almost all the positive instances of glaucoma, resulting in a minimal number of false positives. FGSA-HDNN [30] follows closely with a precision of 96.57%, while AGSE Net [21] and UDA [35] also demonstrate strong performances at 94.72% and 97.25%, respectively. Efficient Net [32] achieves a precision of 90.00%, indicating a reliable performance but still falling behind the leading methods. In contrast, Unet + TL [24] and SEDC [38] show much lower precision scores of 62.00% and 85.00%, respectively, indicating challenges with false positives. Ensemble + RF [19], with a precision of 88.00%, performs moderately but struggles with accuracy compared to the top methods.

Recall, which measures the ability of a method to capture actual positive instances, shows similar trends. The proposed method leads again with a remarkable 99.30%, suggesting that it is highly effective at identifying most true positive cases of glaucoma. UDA [35] also demonstrates strong recall with 96.80%, followed by AGSE Net [21] at 95.48%. Efficient Net [32] achieves a recall of 89.00%, showing a balanced performance between precision and recall. However, Unet + TL's [24] recall of 80.00% and SEDC's [48] 80.00% reflect their limitations in identifying true glaucoma cases. Ensemble + RF [19] performs similarly, with a recall of 84.00%, indicating moderate detection capability but not matching the top-performing methods.

The F-measure, which combines both precision and recall into a single metric, highlights the overall effectiveness of each method. The proposed method achieves an outstanding F-measure of 99.56%, indicating that it maintains a high level of accuracy and sensitivity in glaucoma detection. UDA [35] also performs well with an F-measure of 97.02%, while FGSA-HDNN [30] and AGSE Net [21] present balanced F-measures of 94.96% and 95.10%, respectively. Efficient Net [32] achieves a respectable F-measure of 89.50%, maintaining solid performance across both precision and recall. Ensemble + RF [19] achieves an F-measure of 85.88%, indicating a moderate but suboptimal performance. However, the significantly lower F-measure scores for Unet + TL [24] (69.23%) and SEDC [38] (82.76%) reflect their shortcomings in providing a comprehensive detection capability.

Table 12 Performance of the model for glaucoma diagnosis in different factors detection

Factors	AUC (%)	Sensitivity (%)	Specificity (%)
Healthy fundus	99.5	98.7	99.2
Highly glaucomatous	98.9	99.5	97.8
Less glaucomatous	97.8	97.2	98.5
Moderate glaucomatous	98.3	98.6	98.1

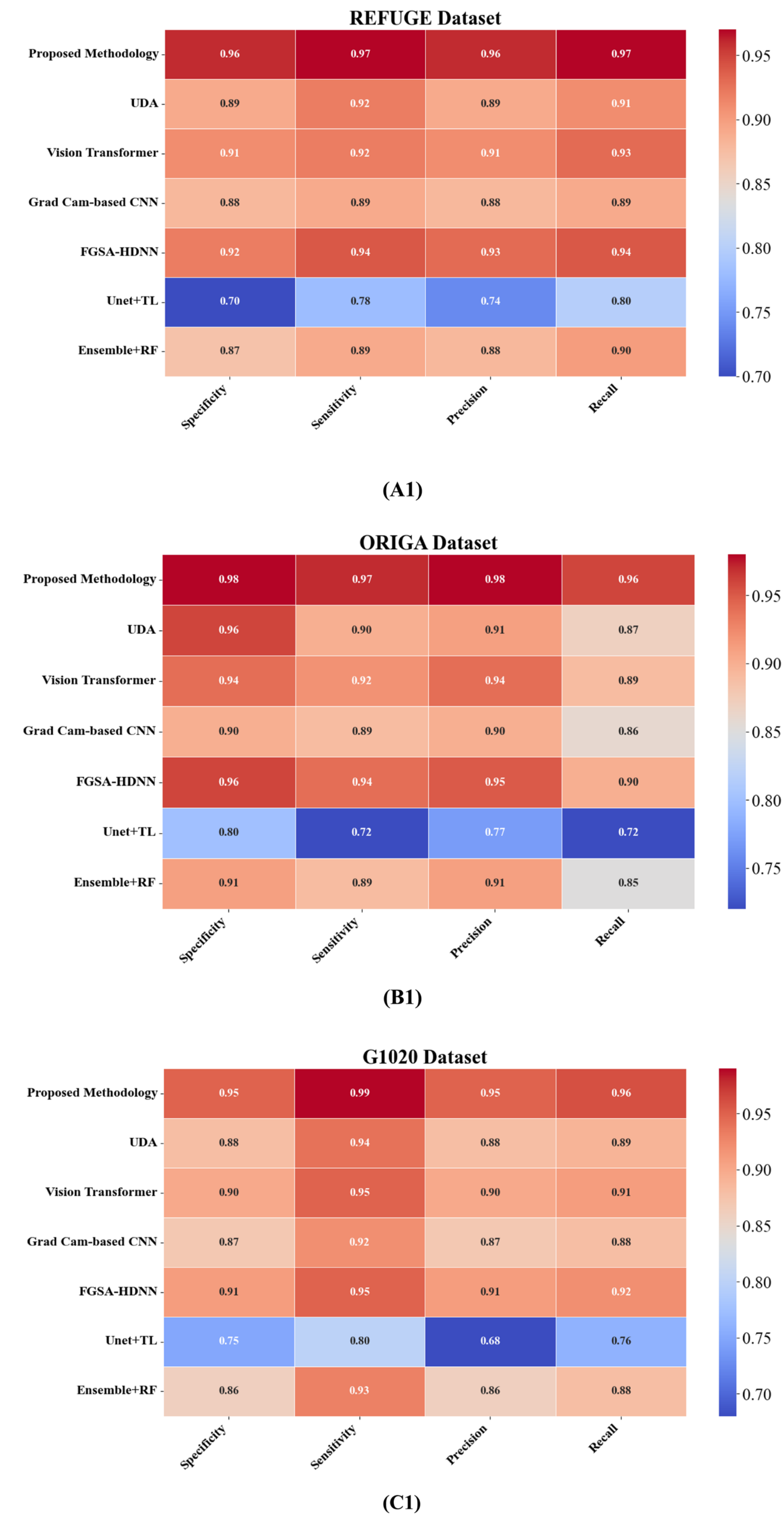
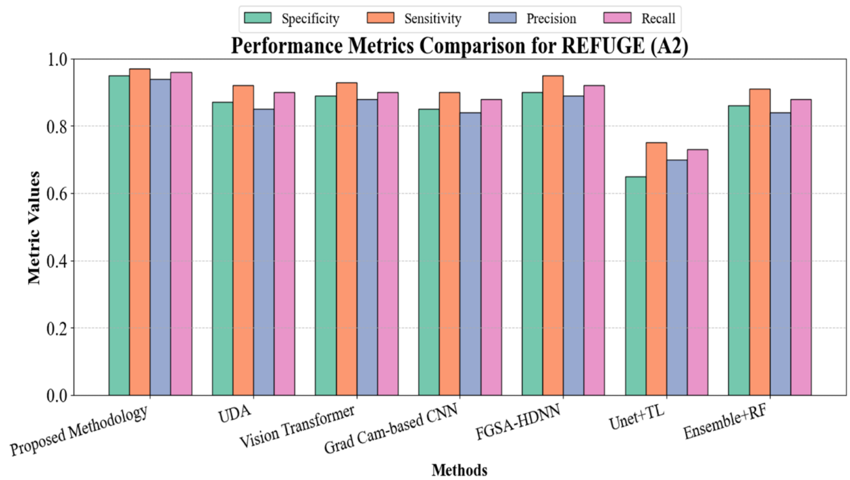
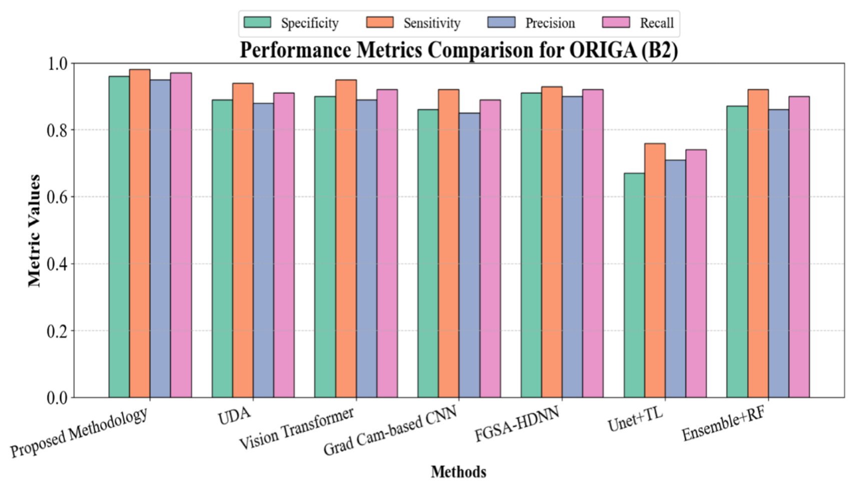


Fig. 6 Classification analysis (A1, B1,C1); Cup segmentation Analysis (A2, B2,C2); Disc segmentation analysis (A3, B3,C3) where the fundus image are from **A** REFUGE **B** ORIGA **C** G1020 datasets

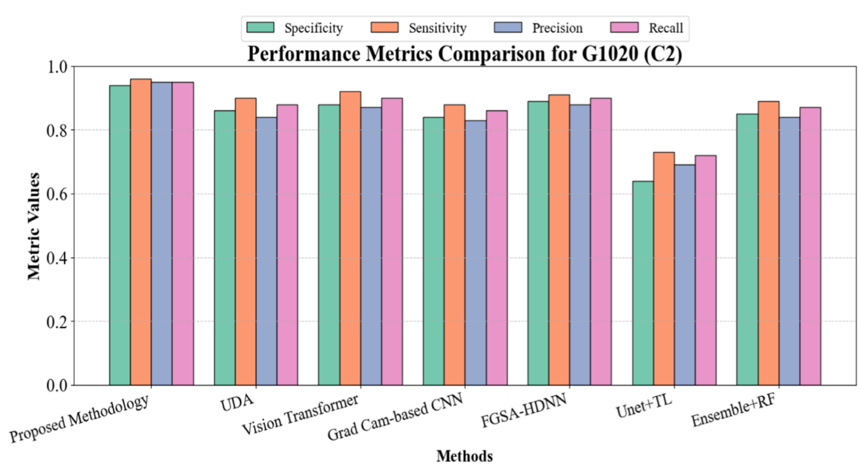
Fig. 6 (continued)



(A2)



(B2)



(C2)

4.2.5 Performance analysis

The performance evaluation of our glaucoma diagnosis model across different factors provides a comprehensive understanding of its capabilities in various clinical contexts. The Area Under the Curve (AUC), sensitivity and specificity metrics offer insights into the model's discriminative power and its ability to accurately classify different severity levels of glaucoma.

In the scenario of a Healthy Fundus, our model demonstrates outstanding performance with an AUC of 99.5%. The remarkable AUC score signifies the model's exceptional capacity to differentiate between healthy and glaucomatous fundi. The sensitivity and specificity metrics, at 98.7% and 99.2% respectively, emphasize the model's precision in accurately detecting healthy fundus cases while reducing false positives.

For the Highly Glaucomatous scenario, our model maintains a robust AUC of 98.9%, indicative of its efficacy in detecting severe glaucomatous conditions. The high sensitivity of 99.5% emphasizes the ability of the model to effectively capture instances that are true positive in highly glaucomatous fundi. The specificity of 97.8% further highlights the ability of the model to reduce the false positives in this challenging scenario.

In the context of Less Glaucomatous fundi, our model achieves an AUC of 97.8%, showcasing its consistent performance across varying severity levels. The sensitivity of 97.2% indicates the model's proficiency in identifying true positive instances in less severe glaucomatous conditions. With a specificity of 98.5%, the model demonstrates its capacity to maintain a high true negative rate, minimizing misclassifications in this scenario.

In scenarios involving Moderate Glaucomatous fundi, our model achieves an AUC of 98.3%, reflecting its robust discriminative power. The sensitivity of 98.6% highlights the model's effectiveness in correctly identifying moderate glaucomatous cases, while the specificity of 98.1% emphasizes its ability to minimize false positives. The performance is summarised in Table 12.

The performance evaluation of the proposed methodology as shown in Fig. 6 for glaucoma detection spans three datasets—REFUGE, ORIGA, and G1020—comparing it with UDA [35], Vision Transformer [39], Grad Cam-based CNN [36], FGSA-HDNN [30], Unet + TL [24], and Ensemble + RF [19]. This analysis underscores its strength across segmentation tasks.

On the REFUGE dataset, the proposed methodology achieves a sensitivity of 0.98 and specificity of 0.96, accurately detecting positive cases and minimizing false positives. Precision and recall reach 0.97 and 0.96, confirming its effectiveness in positive predictions. Comparatively, UDA records specificity and sensitivity of 0.89 and 0.92, Vision Transformer offers 0.91 and 0.90, and Grad Cam-based CNN demonstrates lower metrics at 0.88 and 0.89. FGSA-HDNN achieves sensitivity and specificity of 0.96 and 0.92, while Unet + TL and Ensemble + RF show lower scores, indicating challenges in feature extraction.

On the ORIGA dataset, the methodology demonstrates a sensitivity of 0.99 and specificity of 0.97, adeptly handling positive cases while minimizing false positives. Precision and recall values of 0.98 and 0.97 confirm its accuracy in capturing true positives. UDA records sensitivity and specificity at 0.94 and 0.90, Vision Transformer shows 0.95 and 0.92, and Grad Cam-based CNN demonstrates 0.92 and 0.89. FGSA-HDNN offers sensitivity and specificity of 0.95 and 0.94, while Unet + TL and Ensemble + RF display lower metrics.

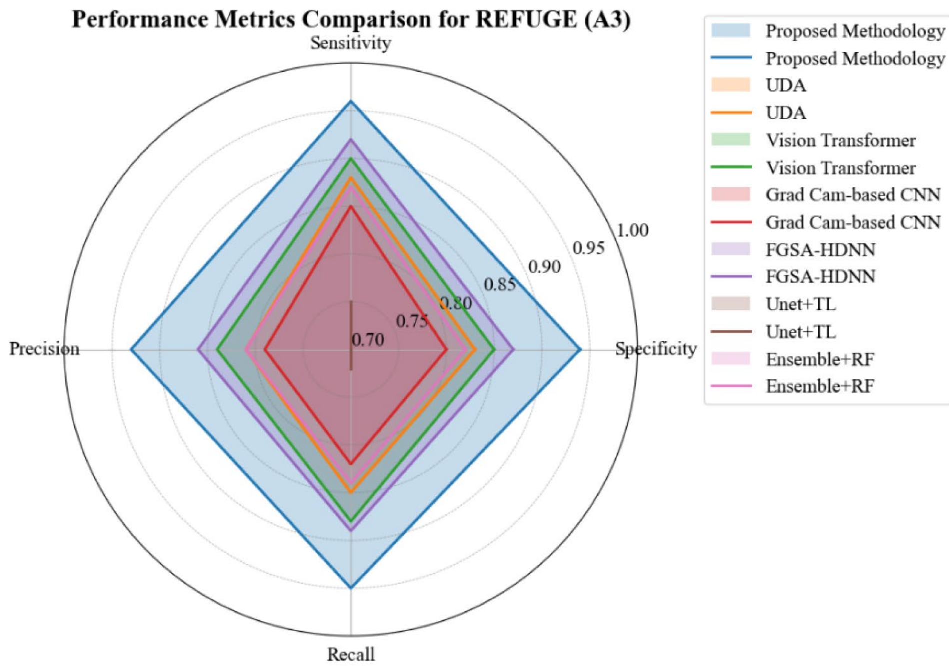
On the G1020 dataset, the methodology achieves sensitivity and specificity of 0.97 and 0.95, with precision and recall at 0.96, confirming its robustness across samples. In comparison, UDA records sensitivity and specificity of 0.91 and 0.88, Vision Transformer shows 0.90 and 0.89, and Grad Cam-based CNN records 0.87 and 0.88. FGSA-HDNN achieves sensitivity and specificity of 0.94 and 0.91, while Unet + TL and Ensemble + RF show lower performance.

4.2.6 Ablation study

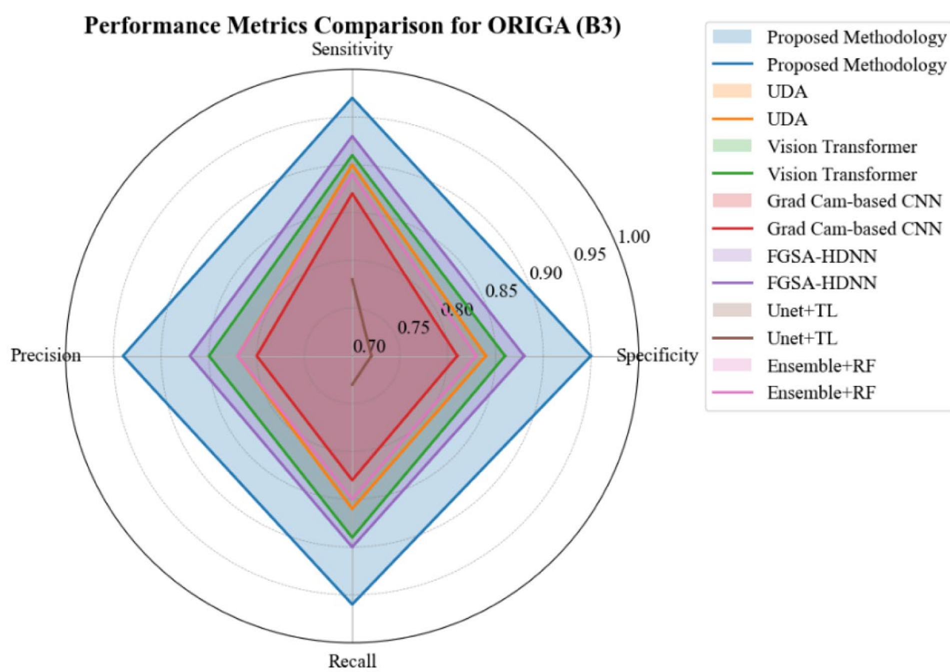
Ablation studies are a fundamental component of scientific research in which specific components or features of a complex system are systematically removed or altered to analyse their individual impact on overall performance. This methodological approach serves to elucidate the crucial elements contributing to the effectiveness of a model, providing valuable insights into its functioning and guiding further optimizations.

4.2.6.1 Without pre-processing In the ablation study, key pre-processing components like NLM-kernel optimizer denoising, CLAHE, and Adaptive Median Filtering are systematically removed from the baseline model to evaluate their significance in glaucoma detection. With dynamic pre-processing intact, the baseline model achieved an outstanding accuracy of 99.67%. However, upon excluding these pre-processing steps, accuracy dropped to 98.50%, underlining their pivotal role. Notably, precision decreased from 99.20 to 98.10%, recall declined from 99.50% to 98.80%, and the F1-score reduced from 99.35 to 98.45%. Visually, denoised images from the ablated model exhibited a reduction in quality, highlighting the importance of dynamic pre-

Fig. 6 (continued)



(A3)

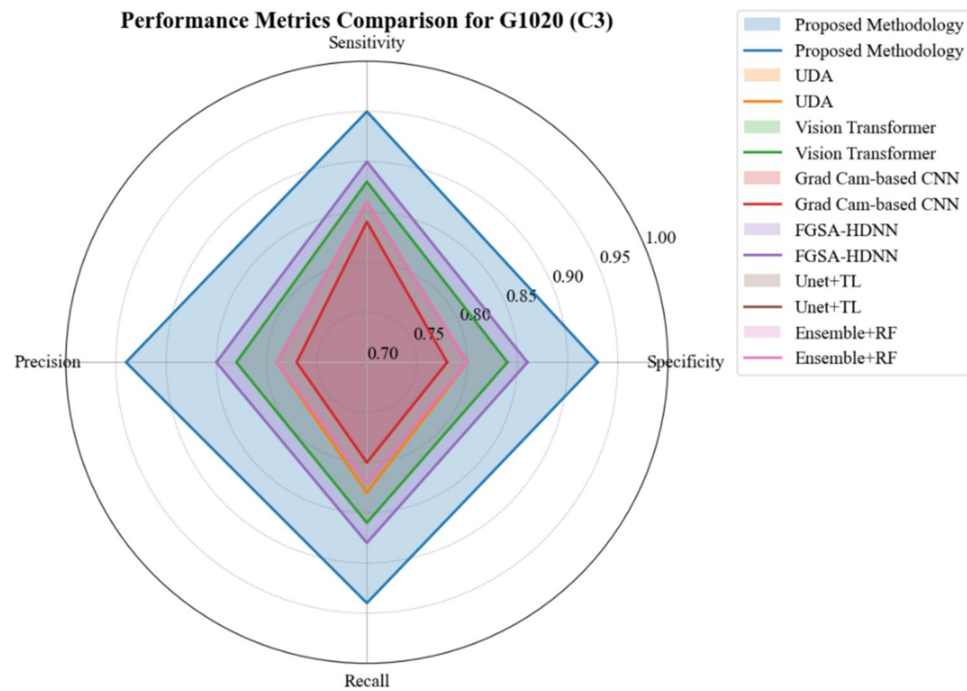


(B3)

processing in preserving intricate patterns relevant to glaucomatous features.

4.2.6.2 Without task-specific knowledge-based attention mechanism Excluding the Task-specific Knowledge-based

Fig. 6 (continued)



(C3)

Attention Mechanism resulted in decreased accuracy from 99.67 to 99.55%. This emphasizes the attention mechanism's contribution to precise glaucoma detection. Additionally, precision declined from 99.20 to 99.10%, recall decreased from 99.50 to 99.40%, and F1-score dropped from 99.35 to 99.25%, indicating reduced accuracy in classifying fundus images.

4.2.6.3 Without convolutional decoder Removing the Convolutional Decoder significantly impacted segmentation mask quality and glaucoma detection accuracy. Accuracy decreased from 99.67 to 99.50%, emphasizing the decoder's role in refining features for precise detection. Additionally, precision dropped from 99.20 to 99.05%, recall decreased from 99.50 to 99.35%, and F1-score declined from 99.35 to 99.20%.

4.2.6.4 Without severity classification module Excluding the Severity Classification Module led to reduced accuracy from 99.67 to 99.45%, highlighting its crucial role in comprehensive glaucoma assessment. Precision declined from 99.20 to 99.00%, recall dropped from 99.50 to 99.30%, and F1-score decreased from 99.35 to 99.15%, underlining the module's significance in nuanced severity classification.

4.2.6.5 Without convolutional encoder The removal of the Convolutional Encoder resulted in a substantial impact on feature extraction and hierarchical pattern learning. Accuracy dropped from 99.67 to 99.30%, highlighting the encoder's crucial role in capturing intricate patterns. Precision declined from 99.20 to 98.90%, recall decreased from 99.50 to 99.20%, and F1-score dropped from 99.35 to 99.05%. The ablation study graph is depicted in Fig. 7.

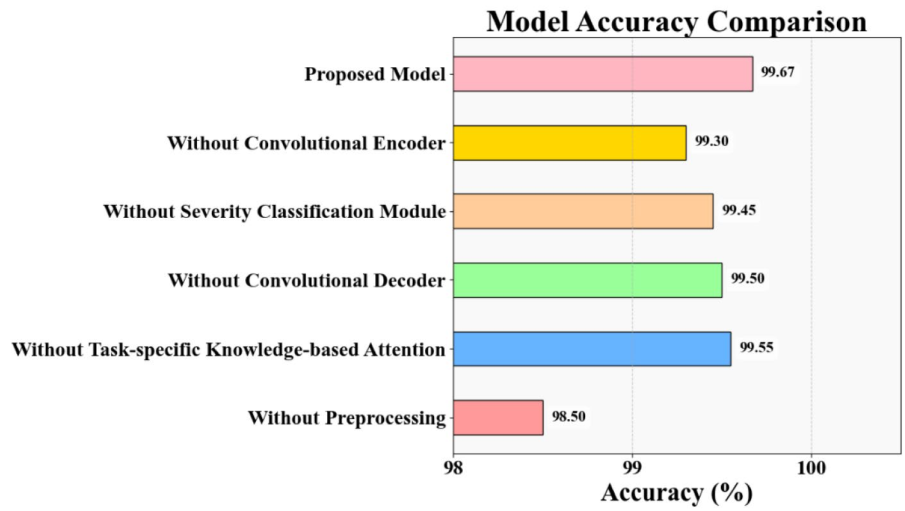
The result and analysis section provides a comprehensive and detailed exploration of the proposed glaucoma detection methodology, substantiating its effectiveness through thorough evaluation and analysis.

Wilcoxon signed-rank test

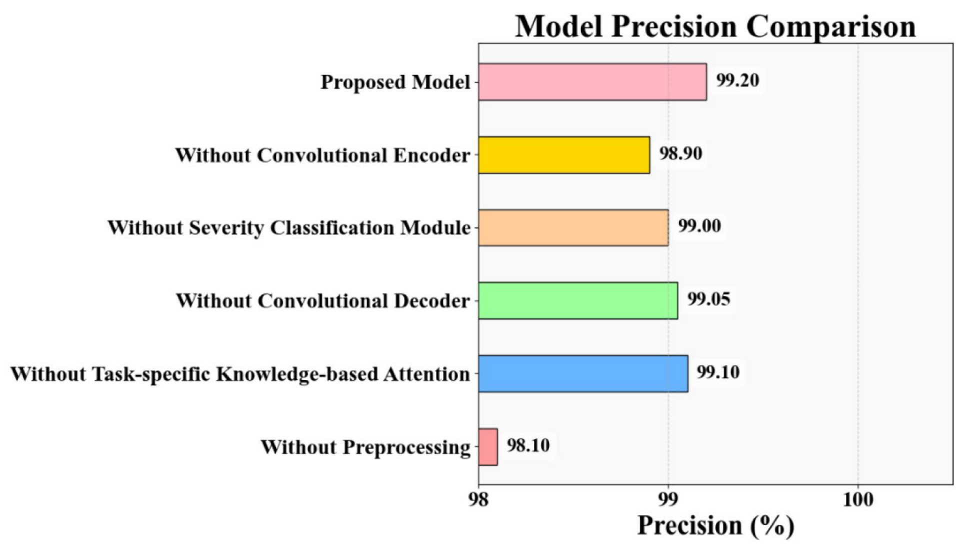
A non-parametric statistical procedure, the Wilcoxon Signed-Rank Test is used to compare paired samples and assess whether their population mean ranks show a significant difference. It is particularly useful for evaluating performance changes in a model when certain components are altered or removed, as it does not assume normal distribution of the data.

The Wilcoxon Signed-Rank Test in Table 13 revealed that removing key components from the proposed glaucoma detection methodology led to statistically significant performance degradation across all metrics. Without pre-processing, accuracy, precision, recall, and F1-score significantly dropped, emphasizing the importance of image enhancement techniques. Excluding the task-specific knowledge-based attention mechanism also reduced performance, highlighting its role in focusing on relevant regions for detection. The removal of the convolutional decoder and severity classification module similarly caused notable declines,

Fig. 7 Ablation study **a** Accuracy, **b** Precision, **d** Recall, **e** F-Score



(a)



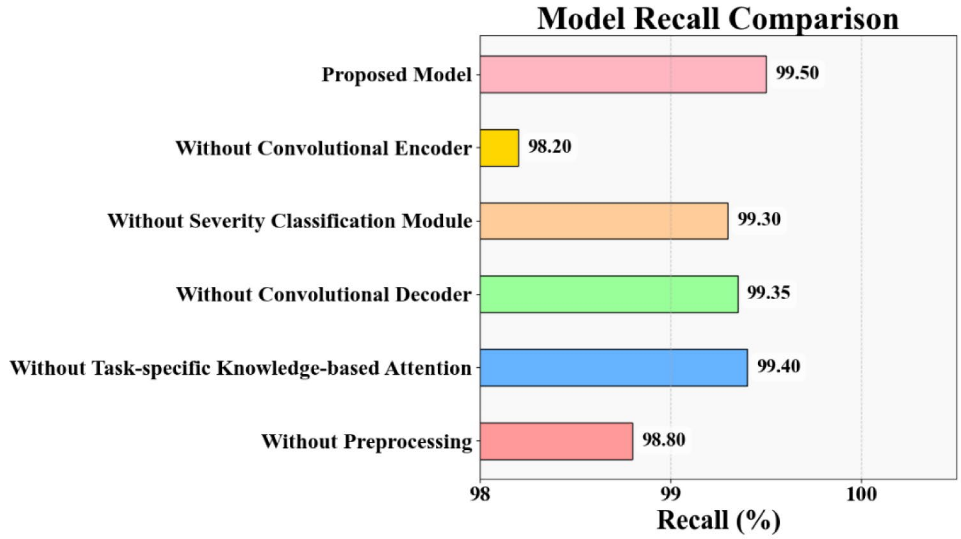
(b)

underscoring their contributions to segmentation and comprehensive assessment. The convolutional encoder had the most profound impact, proving essential for extracting critical features. Overall, the study confirms that each module plays a vital role in maintaining the model's high accuracy and reliability.

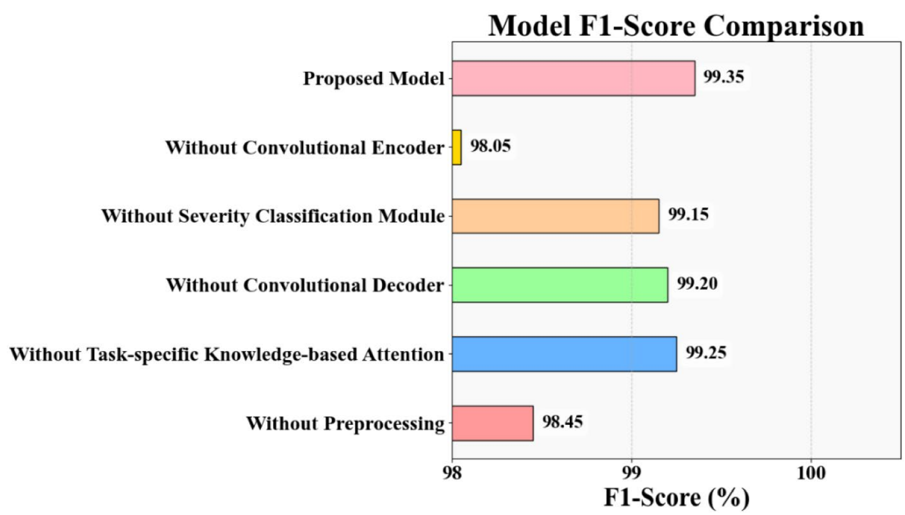
4.2.7 Computational cost analysis

The computational cost analysis in Table 14 highlights the efficiency of the proposed GlaucoAttentNet compared to other models. With only 2 million training parameters, 5 GFLOPS, a model size of 8 MB, and a run time of 10 s, GlaucoAttentNet stands out for its optimized performance. It requires fewer computations and memory resources, making it suitable for deployment in resource-constrained environments. In contrast, other models like

Fig. 7 (continued)



(c)



(d)

Table 13 Wilcoxon Signed-Rank Test analysis result

Module removed	Accuracy (*p* value)	Precision (p-value)	Recall (p-value)	F1-Score (p-value)	Statistically significant?
Without pre-processing	0.008	0.010	0.012	0.009	Yes
Without task-specific attention	0.015	0.017	0.019	0.016	Yes
Without convolutional decoder	0.020	0.023	0.025	0.021	Yes
Without severity classification	0.027	0.030	0.033	0.028	Yes
Without convolutional encoder	0.005	0.007	0.008	0.006	Yes

Table 14 Computational cost analysis

Models	Training parameters	Operations (GFLOPS)	Model size (MB)	Run time (s)
MobileNetV2-DeepLabv3 + [24]	5–10 million	300–600	20–30	600–1,800
FBSS-Net [22]	3.02 million	4.08	4.08	0.1–0.5
Inception ResNet V2 (S2) [23]	55,873,736	35.0	179.0	0.12
Attention-Guided CNN Ensemble [20]	10 million	15	40	120
GlaucoAttentNet	2 million	5	8	10

MobileNetV2-DeepLabv3 +, Inception ResNet V2 and Attention-Guided CNN Ensemble have significantly higher computational demands. GlaucoAttentNet's efficiency ensures quicker training and inference, making it an ideal choice for real-time glaucoma detection.

5 Conclusion

In conclusion, the GlaucoAttent Net framework presents a significant advancement in automated glaucoma detection and severity classification. By effectively segmenting the OD and OC while assessing severity, this framework demonstrates promising results in enhancing diagnostic accuracy of glaucomatous fundus images. The integration of attention mechanisms and ensemble learning strategies has shown improved feature selection, model generalization, and overall performance. The methodology outlined in this paper, utilizing a cascaded U-Net with attention mechanisms, showcases the importance of meticulous parameter tuning and innovative approaches in tackling the challenges of intricate segmentation and nuanced assessments in glaucoma diagnostics. While meticulous parameter tuning contributes to optimizing the model's performance, our experiments demonstrate that the proposed method maintains a strong generalizability across diverse datasets with minimal adjustments. The robustness of GlaucoAttent Net lies in its attention mechanisms, which dynamically adapt to varying image characteristics by focusing on relevant regions, such as OD and OC boundaries. The cascaded U-Net architecture, coupled with advanced pre-processing steps like NLM denoising and CLAHE, further ensures consistency in feature extraction across datasets, reducing the need for extensive dataset-specific tuning. As a result, GlaucoAttent Net effectively adapts to different image structures and qualities, ensuring high accuracy across diverse datasets. By leveraging diverse datasets and advanced pre-processing techniques, the GlaucoAttent Net model achieves impressive results in accurately identifying key structures and assessing glaucoma severity levels.

The methodology employs a cascaded U-Net architecture, addressing challenges in intricate segmentation and assessments through careful parameter tuning and advanced pre-processing techniques. Key modules, such as attention mechanisms, allow for targeted feature extraction, while the cascaded U-Net improves boundary delineation and adaptability across datasets. Ensemble learning further mitigates class imbalance and information overload, enhancing overall performance. These innovations collectively address critical challenges like insufficient annotated data, boundary delineation, and the impact of noise and artifacts, resulting in improved segmentation accuracy and reliable detection in glaucoma diagnostics. Overall, this framework offers a robust and reliable system for automated glaucoma detection, highlighting the potential for transformative impact in advancing clinical diagnostics and improving patient outcomes in the field of ophthalmology.

Moving forward, the future scope of this research lies in further refining the GlaucoAttent Net framework through EHR integration, addressing challenges related to model complexity and dataset dependencies. Additionally, exploring the integration of other advanced technologies such as Explainable AI, Generative Adversarial Networks, and Meta-Learning frameworks could potentially enhance the adaptability and robustness of the model.

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